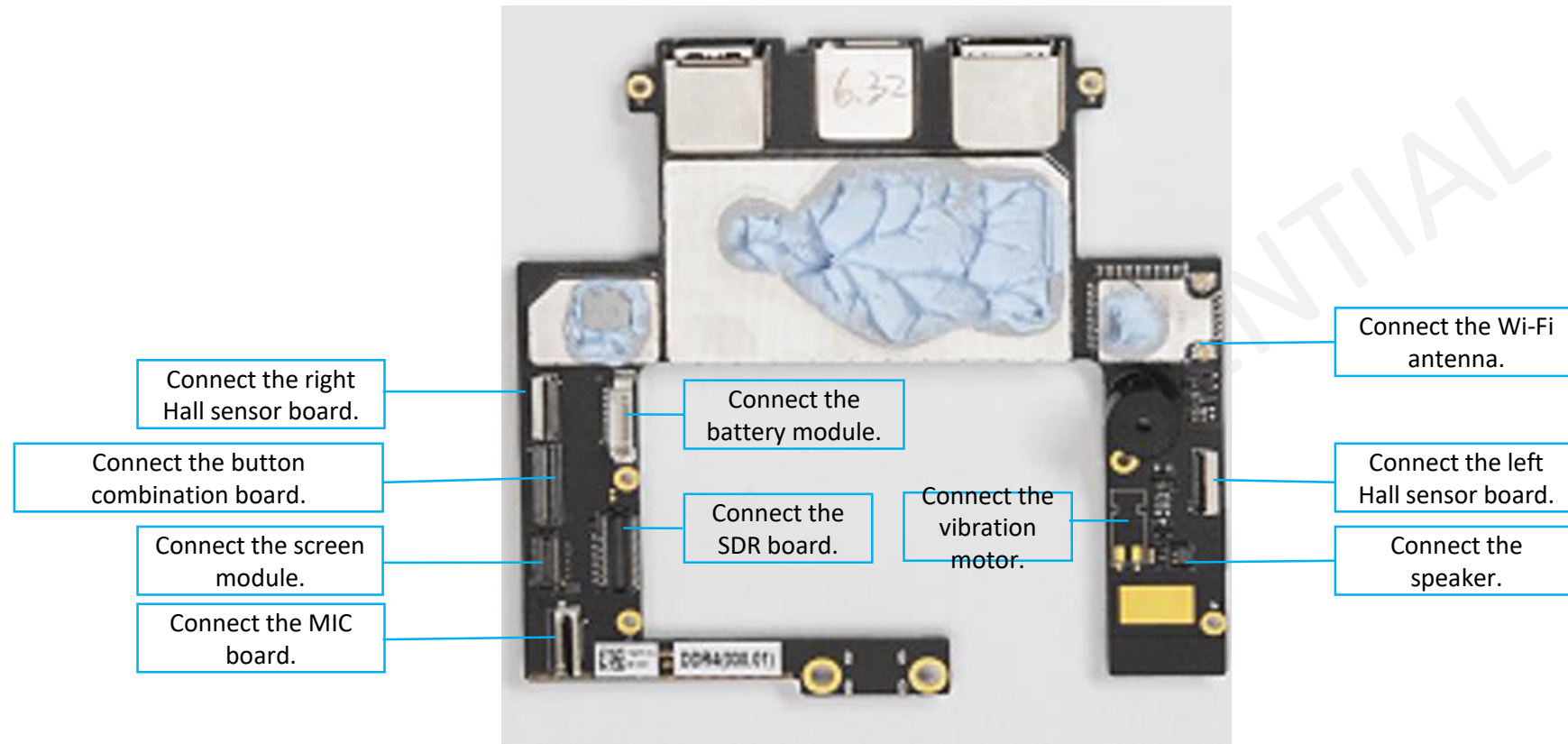
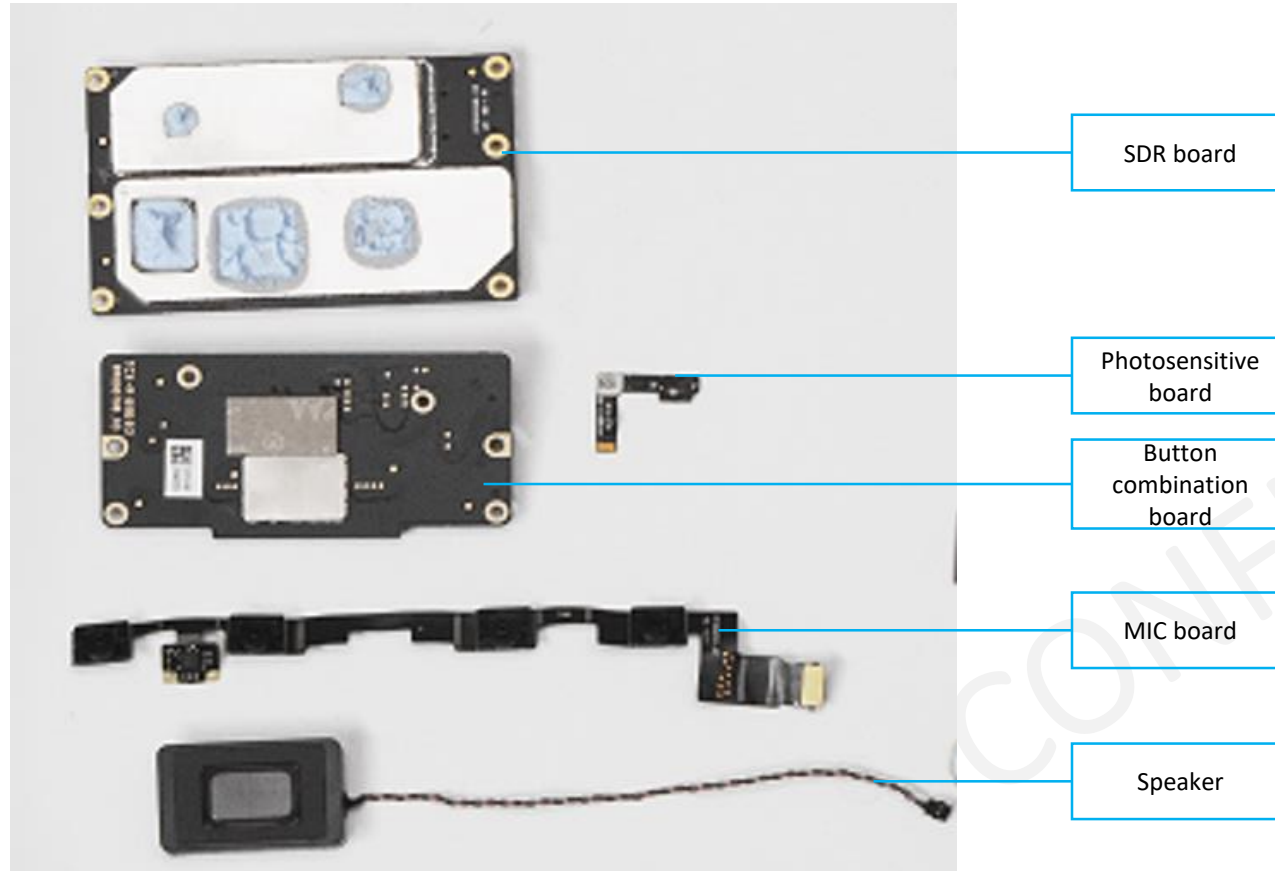




T20 Controller Repair Training



Remote controller main board: Handles sensor information and provides corresponding information to the aircraft and screen module.



SDR board: It is mainly used to handle the image transmission signal and aircraft signal.

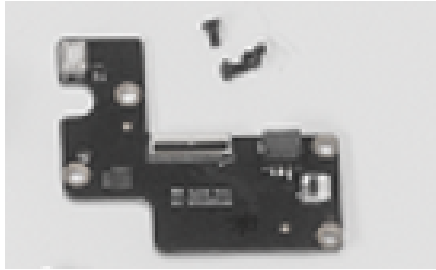
Photosensitive board: It is mainly used to check the ambient light intensity and provide feedback to the remote controller main board for adjusting the screen brightness.

Button combination board: It is mainly used to receive information from the RTH Button, C3 Button, Flight Mode Switch, 5D Button, and Power Button. Receives information from the photosensitive board and provides feedback to the remote controller main board.

The button combination board includes the GPS module.

MIC board: Remote controller speaker.

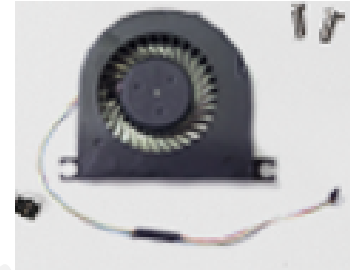
Speaker: Plays sounds from the app.



Hall sensor board



Battery module

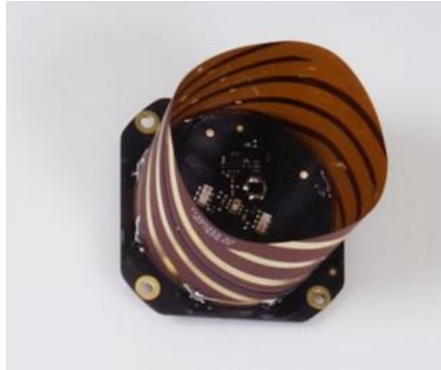


Fan

Hall sensor board: Collects the dial, button and control stick signals, and provides feedback to the remote controller main board.

Battery module: The remote controller's built-in battery is used to provide power supply to each module.

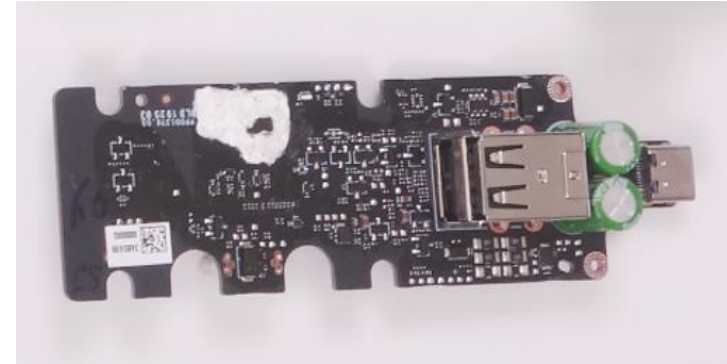
Fan: Cools the SDR board.



RTK antenna
module



Port board of the RTK high-
precision positioning module



Port board of the
expansion module

RTK antenna module: Receives the GPS signal.

Port board of the RTK high-precision positioning module: Handles the GPS signal.

Port board of the expansion module: Connects the dongle and external battery, and transmits the power supply and dongle signal.

RM500 Smart Controller - Troubleshooting



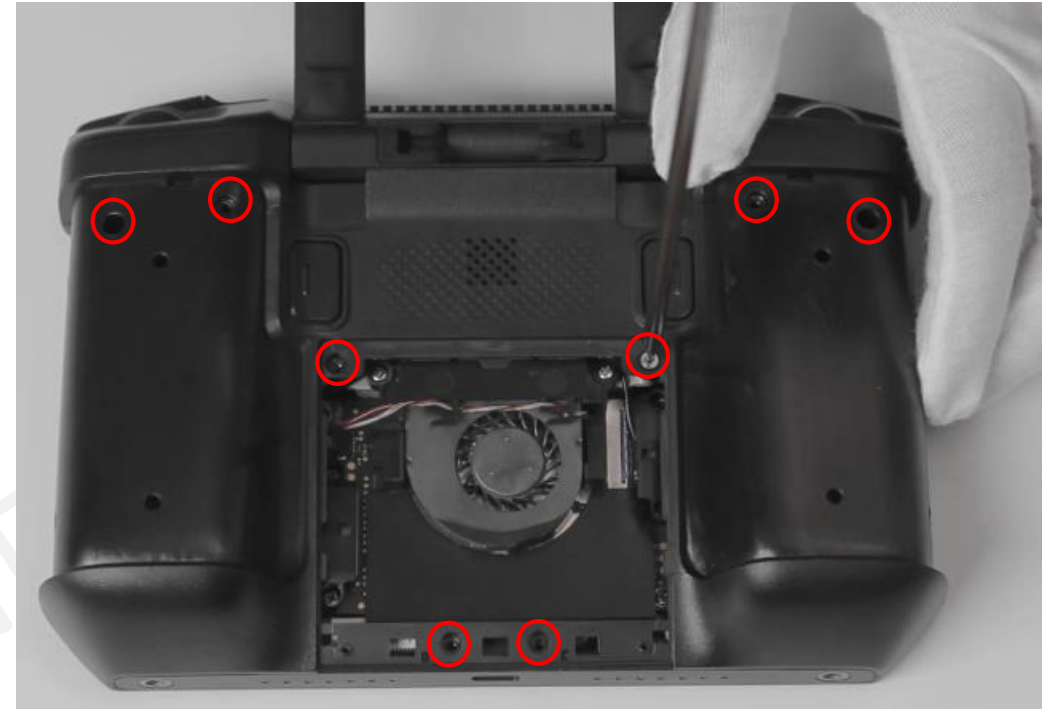
Component	Failure	Reason Analysis	Solution
RM500 Smart controller	Unable to charge	Check whether the built-in battery and remote controller main board are connected properly.	Reinsert and tighten the corresponding cable.
		Check the connection cable between the port board of the expansion module and the remote controller main board.	Reinsert and tighten the corresponding cable.
		Replace the port board of the expansion module for cross testing.	Replace the port board of the expansion module.
		Replace the built-in battery for cross testing.	Replace the built-in battery.
		Replace the remote controller main board for cross testing.	Replace the remote controller main board.
	Unable to power on	Check the connection link of the button combination board, main board, and battery.	Reinsert and tighten the corresponding cable.
		Replace the button combinations board for cross testing.	Replace the button combination board.
		Perform a cross test for the built-in battery.	Replace the built-in battery.
		Replace the main board for cross testing.	Replace the remote controller main board.
	The aircraft cannot be linked.	Check the firmware version of the remote controller and aircraft.	Update the firmware of the remote controller and aircraft to the latest version.
		Replace the SDR board for cross testing.	Replace the SDR board.
		Replace the remote controller main board for cross testing.	Replace the remote controller main board.
	The remote controller satellite search is abnormal.	Replace the button combination board for cross testing.	Replace the button combination board.
	The screen is abnormal.	Check whether the screen or the coaxial cable is damaged.	Replace the front shell component (including screen).
	The Wi-Fi cannot be connected.	Check whether the Wi-Fi antenna is damaged or connected properly.	Reinsert and tighten the Wi-Fi antenna of the corresponding cable.
		Replace the remote controller main board for cross testing.	Replace the main board
	The dial or button is malfunctioning.	Check whether the dial or button is obstructed.	Replace the corresponding dial and button.
		Check the cable that connects the dial and button and the button combination board or the remote controller main board.	Reinsert or tighten the connection cable.
		Replace the button combination board or Hall sensor board for cross testing.	Replace the button combination board or Hall sensor board for cross testing.
		Replace the remote controller main board for cross testing.	Replace the remote controller main board.
Expansion module	The battery cannot be connected.	The type-C port of the remote controller main board is damaged.	Replace the main board
		The Type-C port of the expansion module is damaged.	Replace the port board of the expansion module.
	The 4G dongle cannot be connected.	The type-C port of the remote controller main board is damaged.	Replace the main board
		The Type-C port of the expansion module is damaged.	Replace the port board of the expansion module.
RTK Dongle	The RTK satellite search is abnormal.	Check the RTK port.	Tighten or replace the RTK board.
		Check the remote controller main board port.	Replace the remote controller main board.

RM500 Smart Controller Disassembly

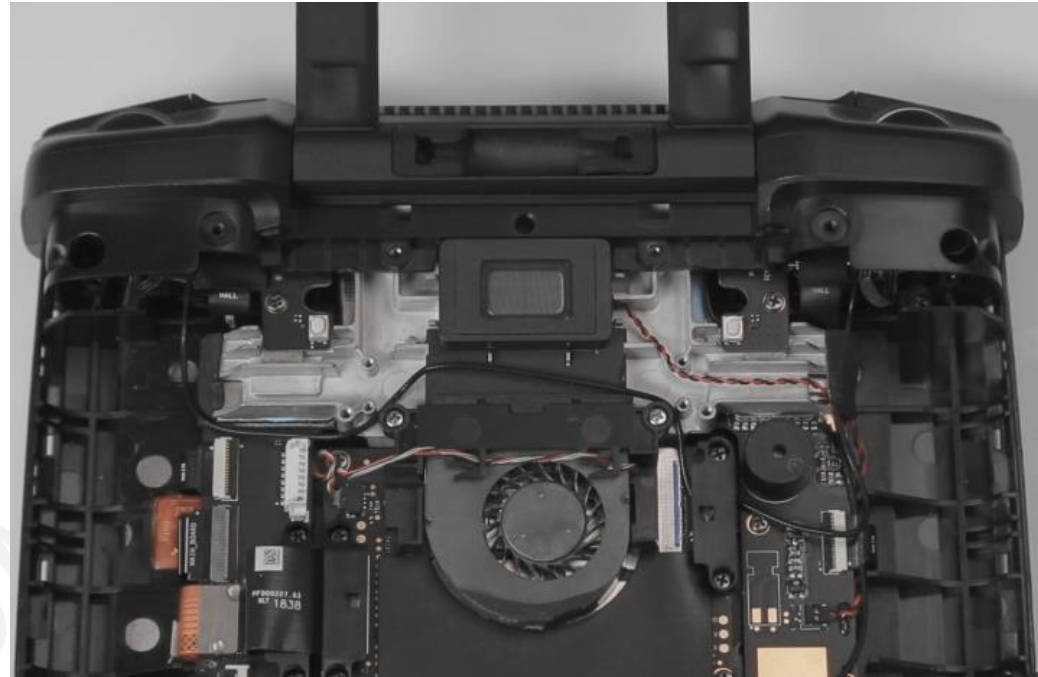
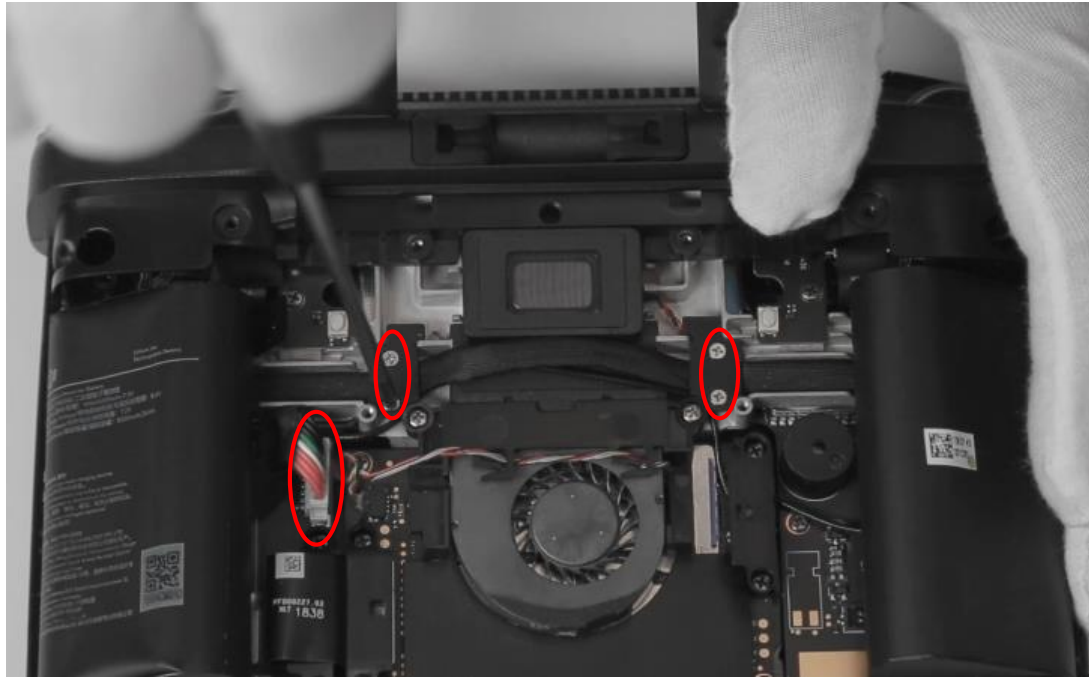


Detach the RTK dongle module vertically.

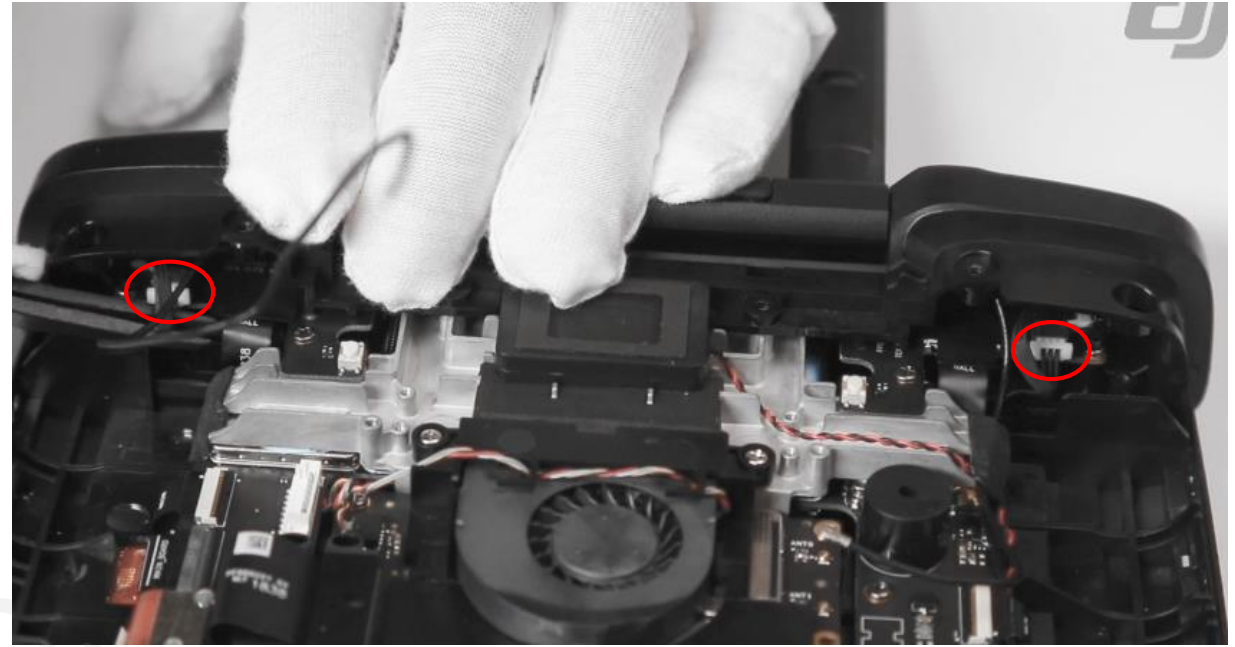
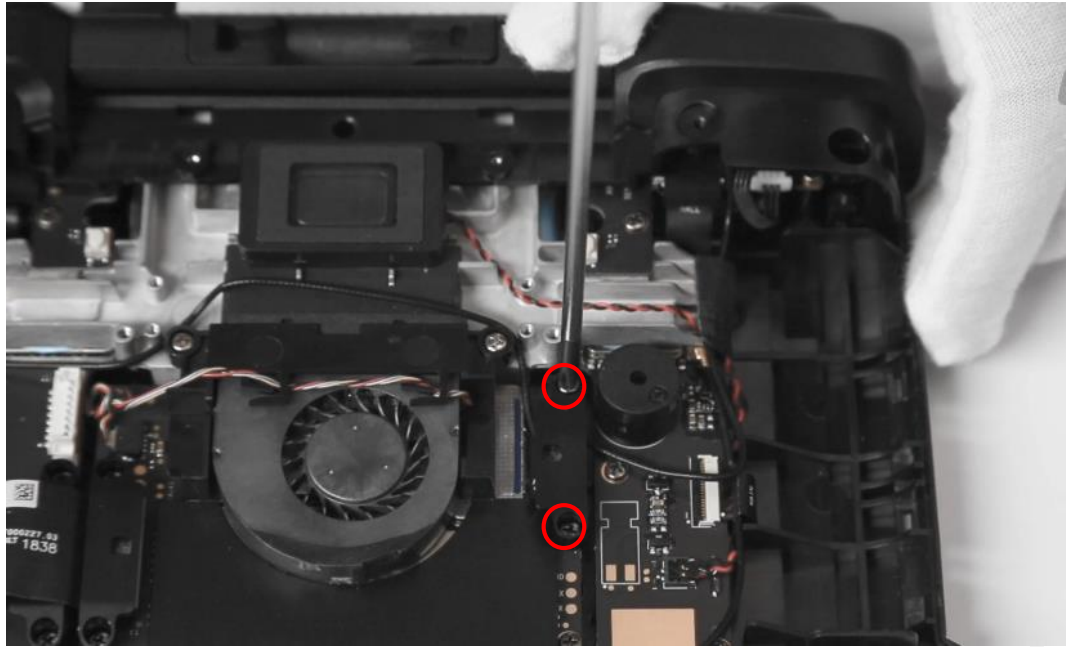
Remove the remote controller expansion module. Note that the back side of the expansion module is applied with double-sided adhesive tape. Please carefully apply force.



Unscrew the two screws securing the rear cover, then remove the rear cover.
Peel off the left and right rubber pads. Unscrew the eight screws securing the lower shell,
then remove the lower shell.

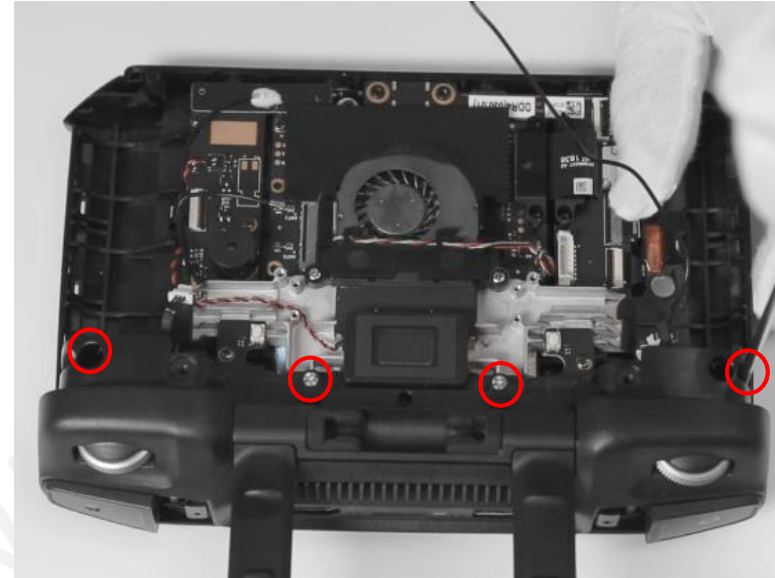
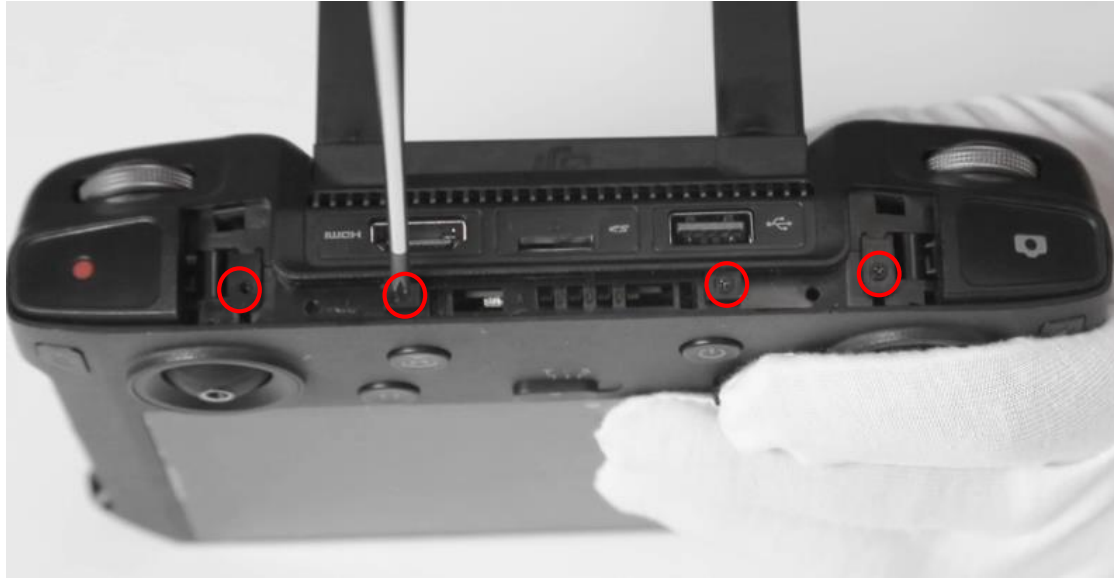


Unscrew the four screws securing the two battery cable mounting pieces, and then remove the battery cable mounting pieces.
Pull out the battery plug then remove the battery.



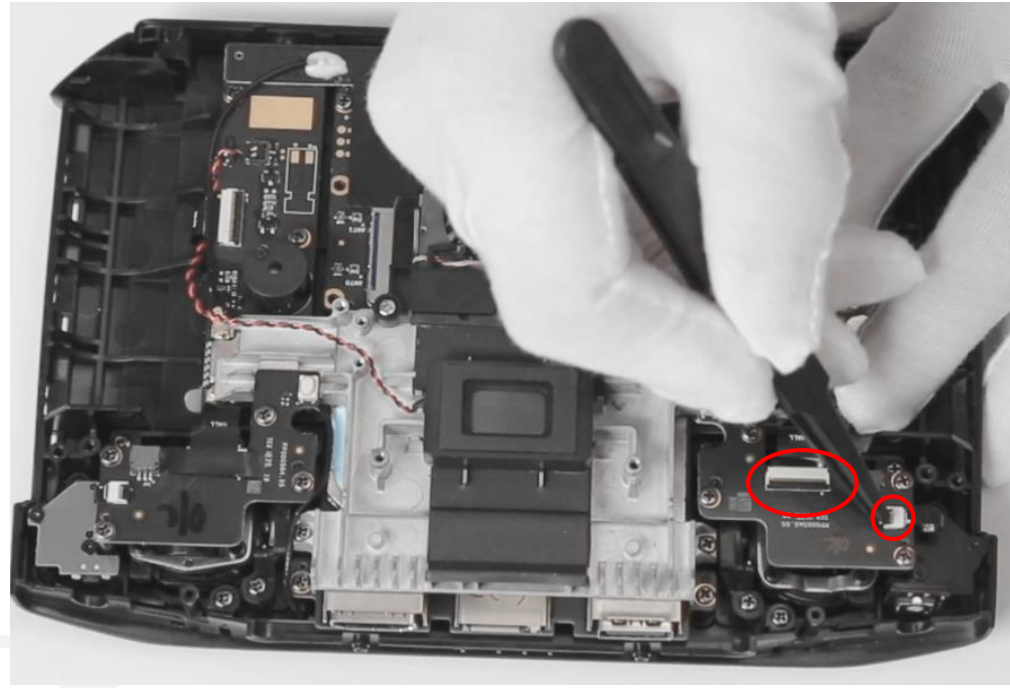
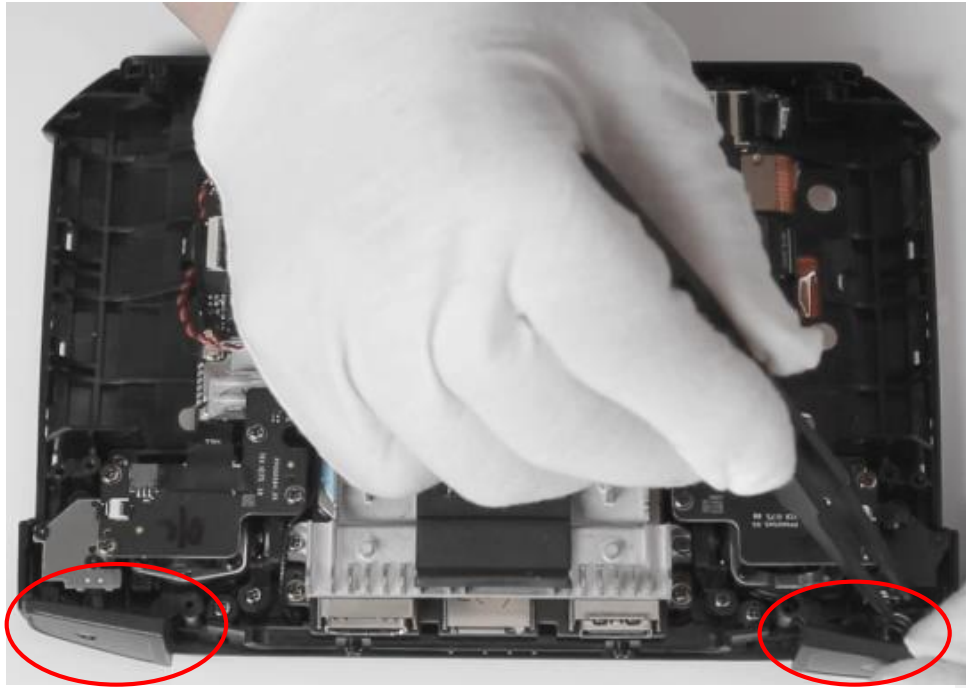
Unscrew the two screws securing the image transmission antenna mounting piece, remove the image transmissi antenna mounting piece, and then disconnect the image transmissi antenna buckles.

Pull out the left and right dials' flat cables.

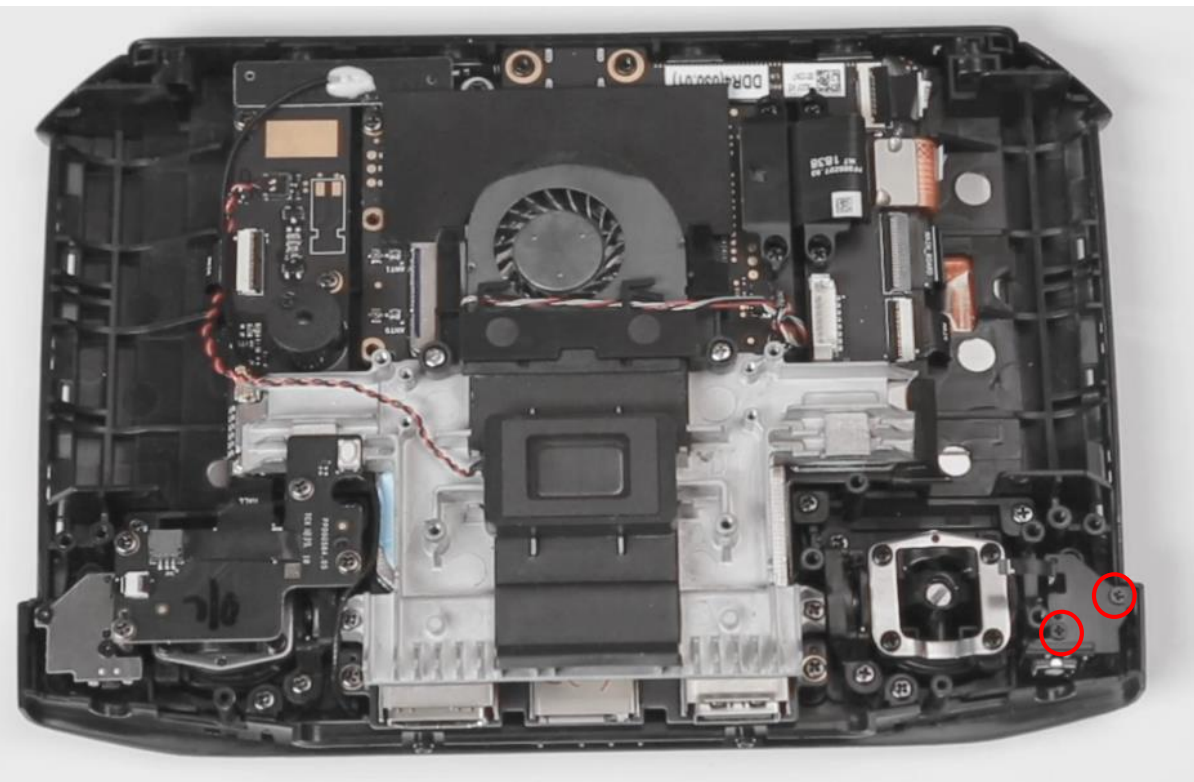
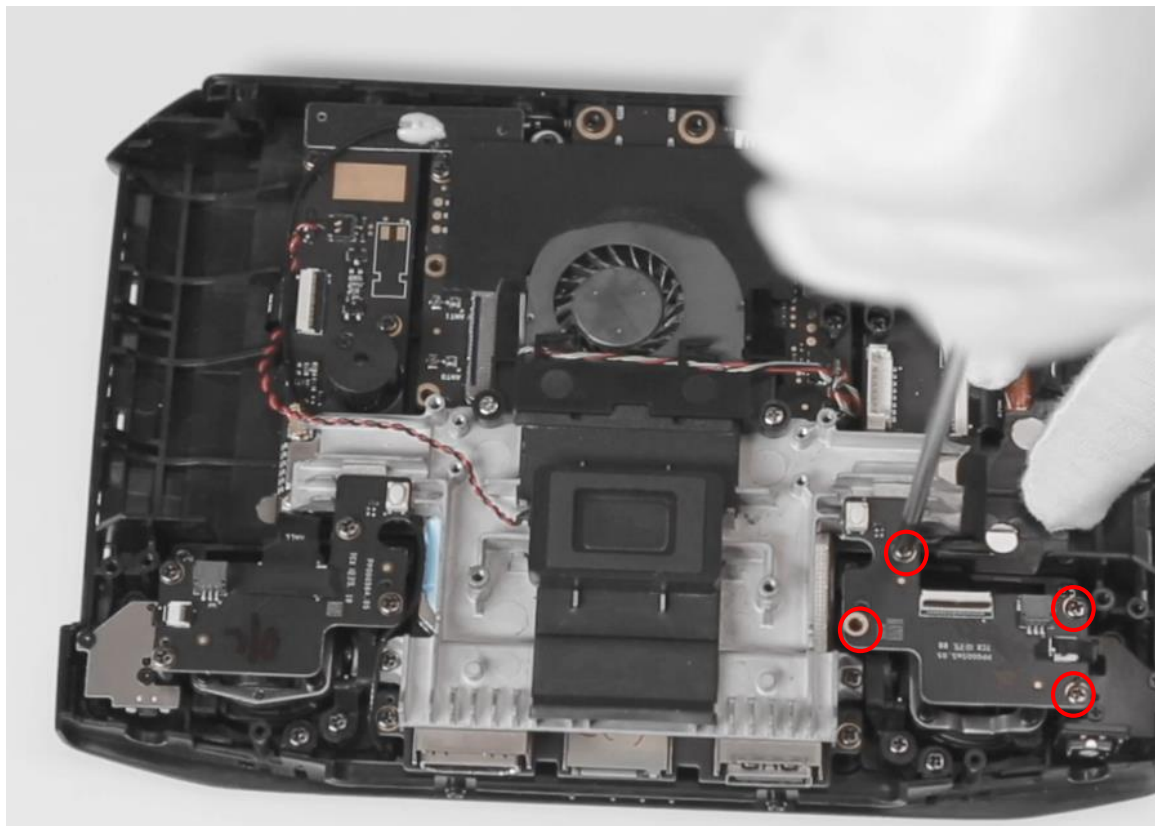


Remove the front decorative cover.

Unscrew the eight screws securing the front cover module, and then remove the front cover module.

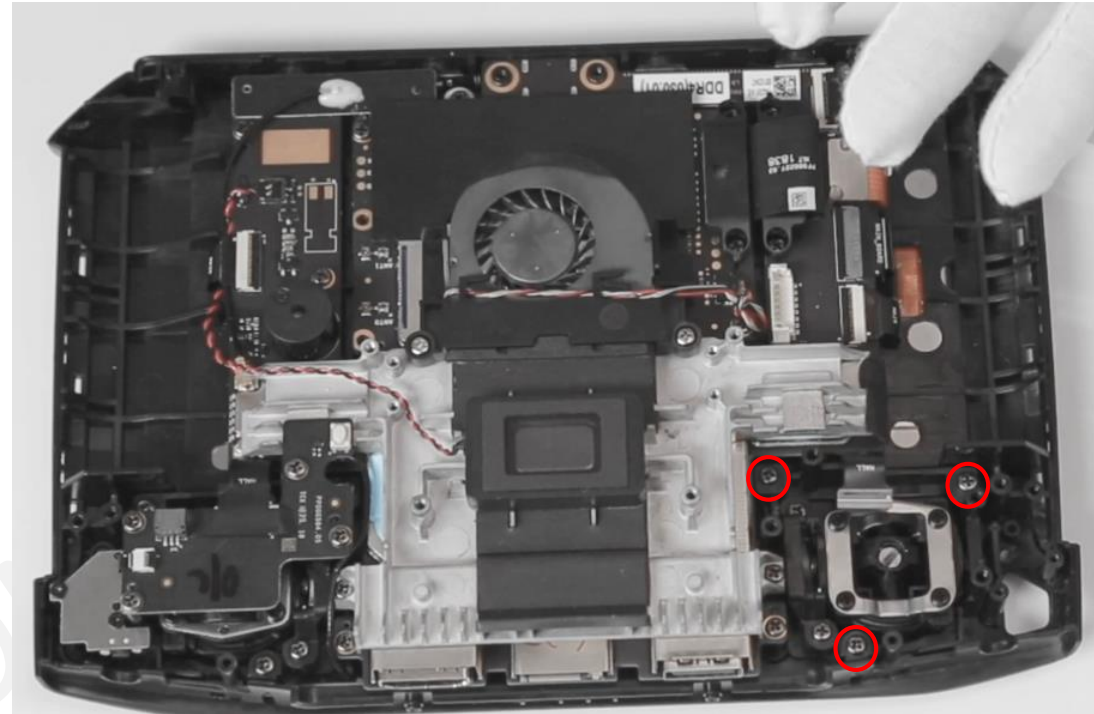
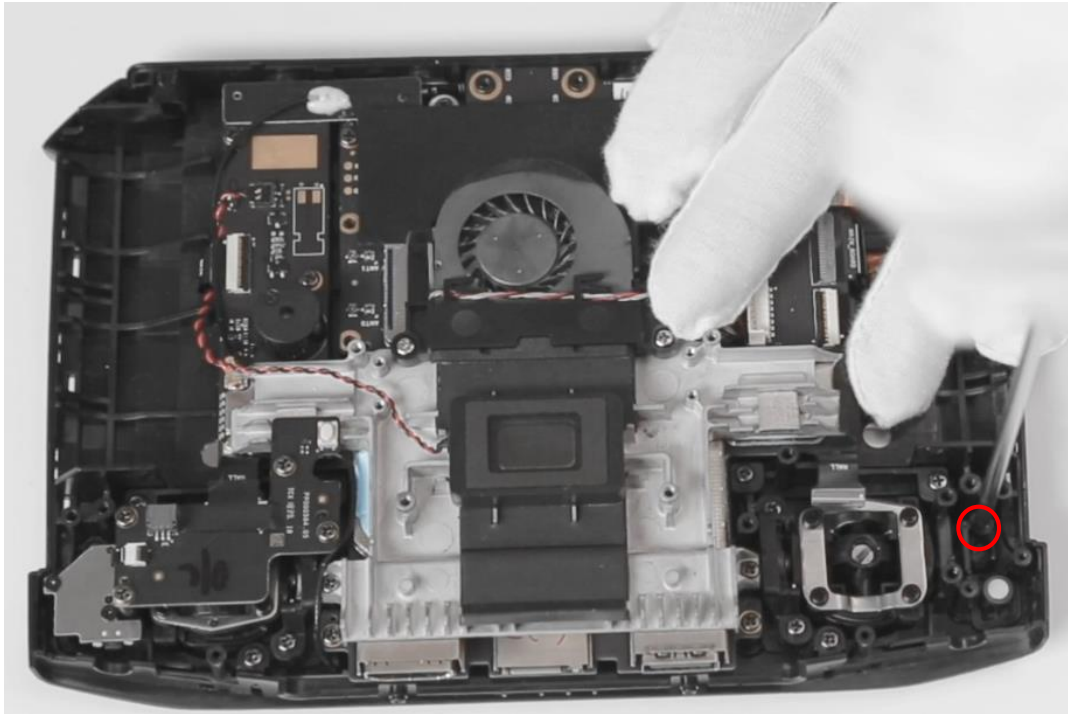


Remove the Spray Button, FPV/Map Switch Button and spring.
Disconnect the button board flat cable and the flat cable that connects the main board with the right Hall sensor board.

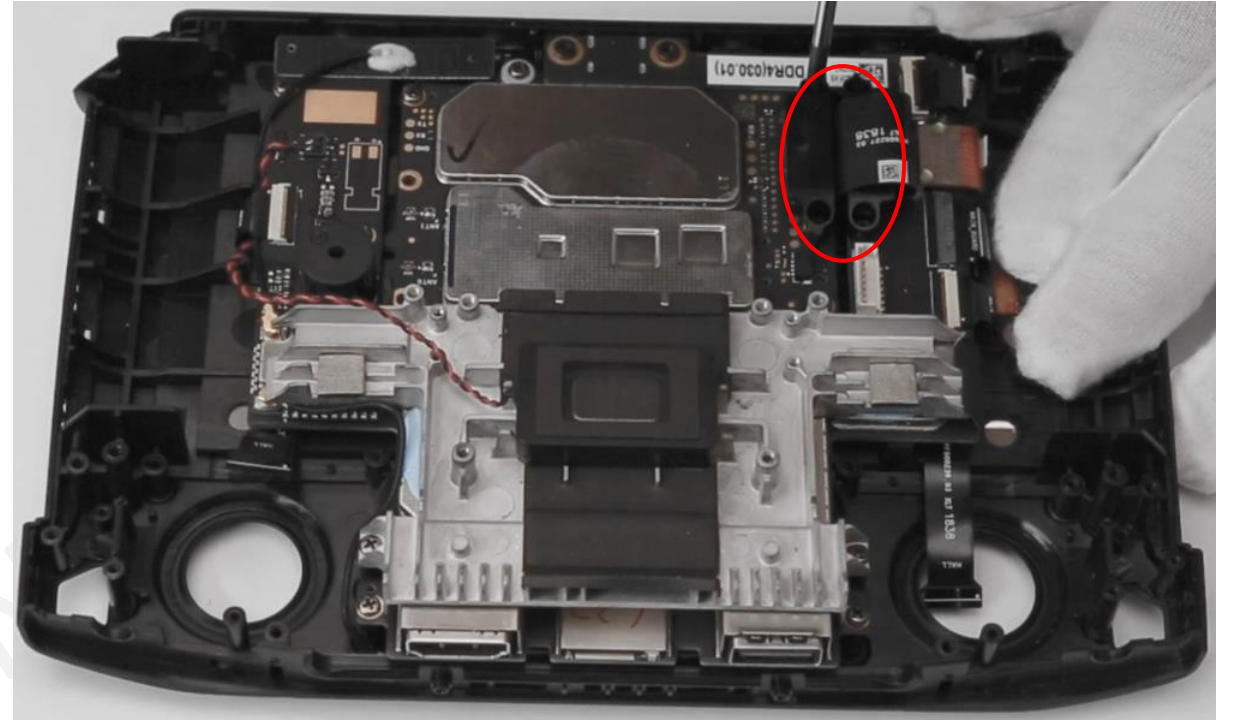
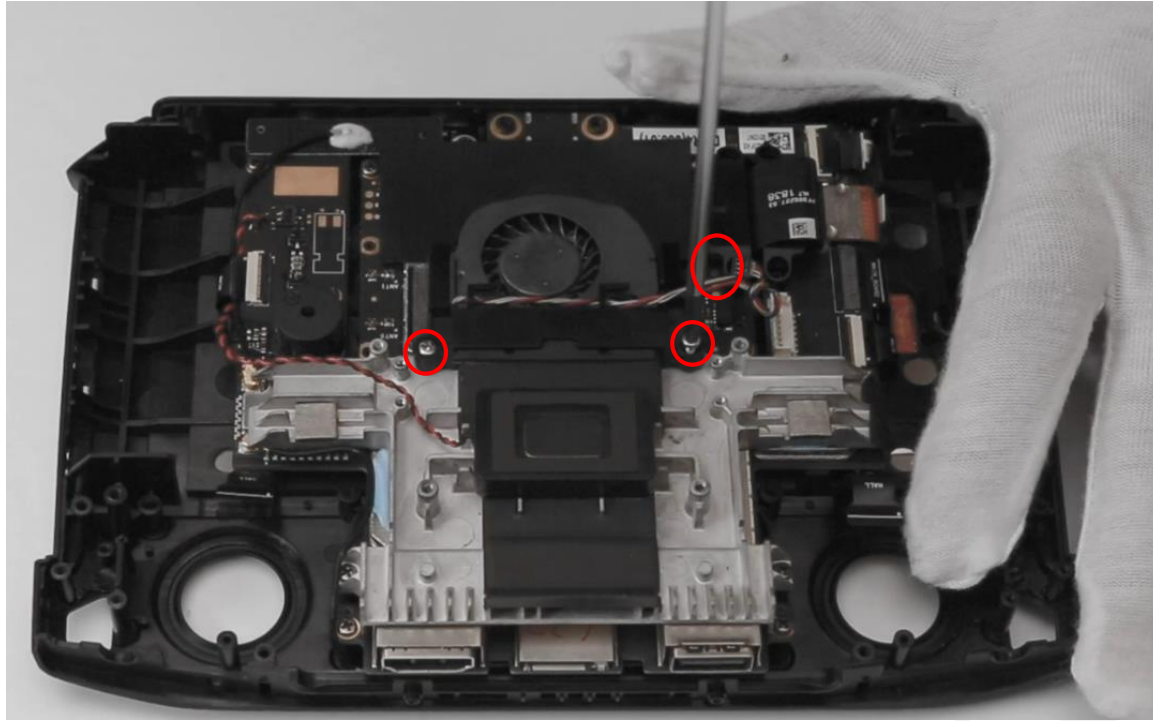


Unscrew the four screws securing the right Hall sensor board, and then remove the right Hall sensor board.

Unscrew the two screws securing the button board, and then remove the FPV/Map Switch button board.

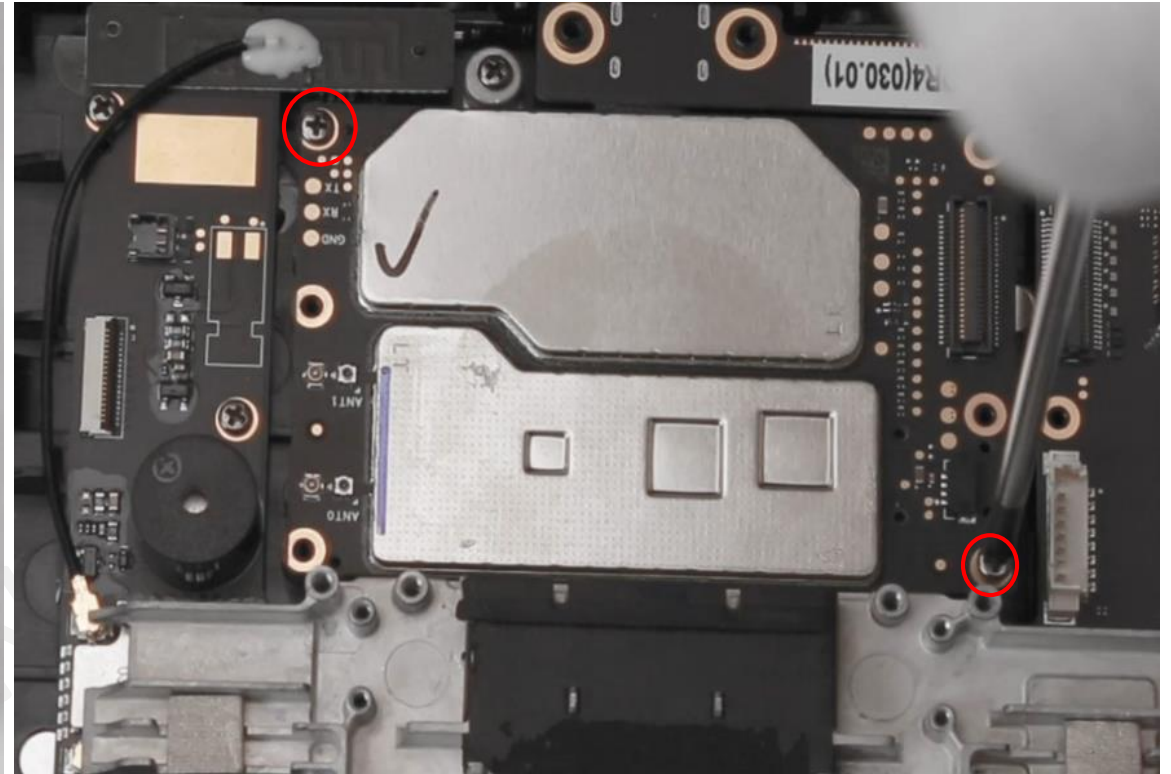
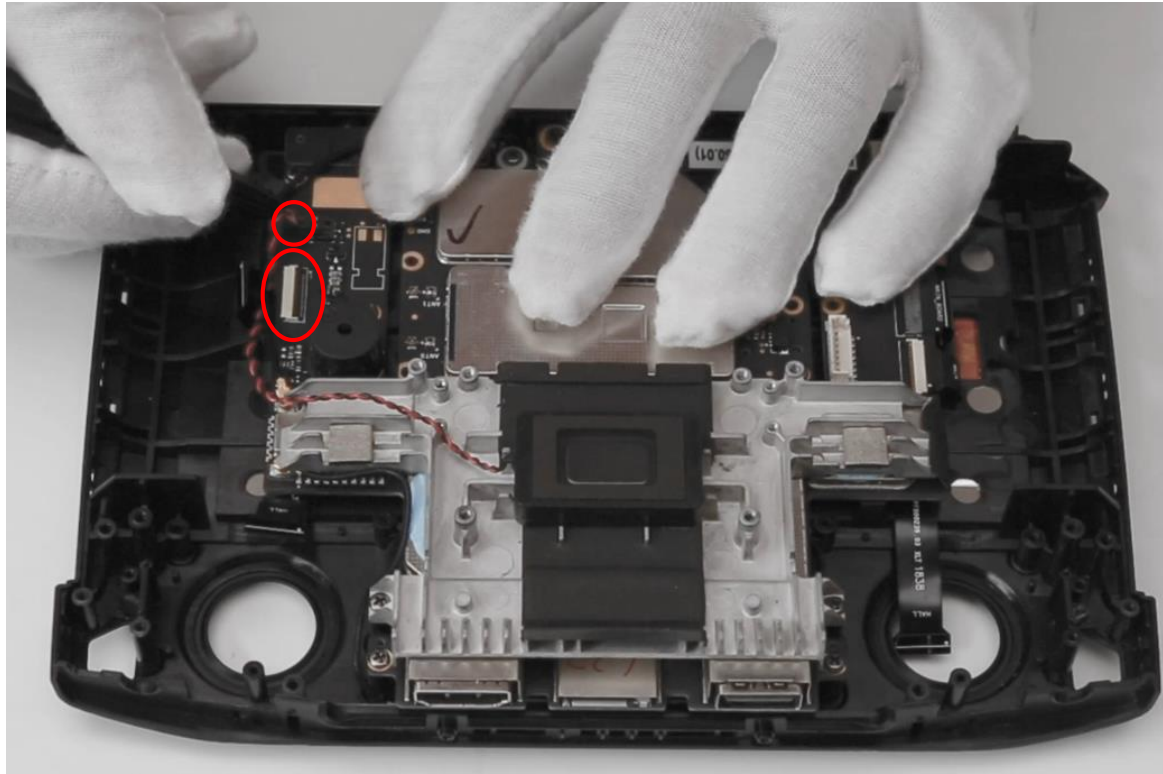


Unscrew the screw that secures the Confirm Button, and then remove the Confirm Button.
Unscrew the three screws securing the right control stick module, and then remove the right control stick module.
Remove the left Hall sensor board, left control stick module, Spray Button Board, Back Button/Function Button.



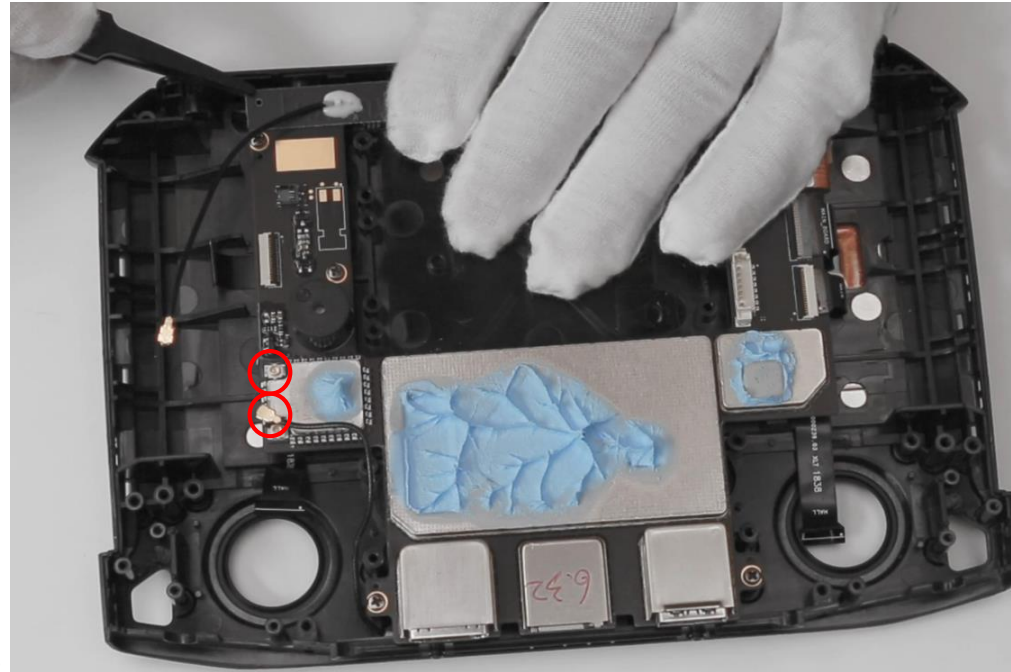
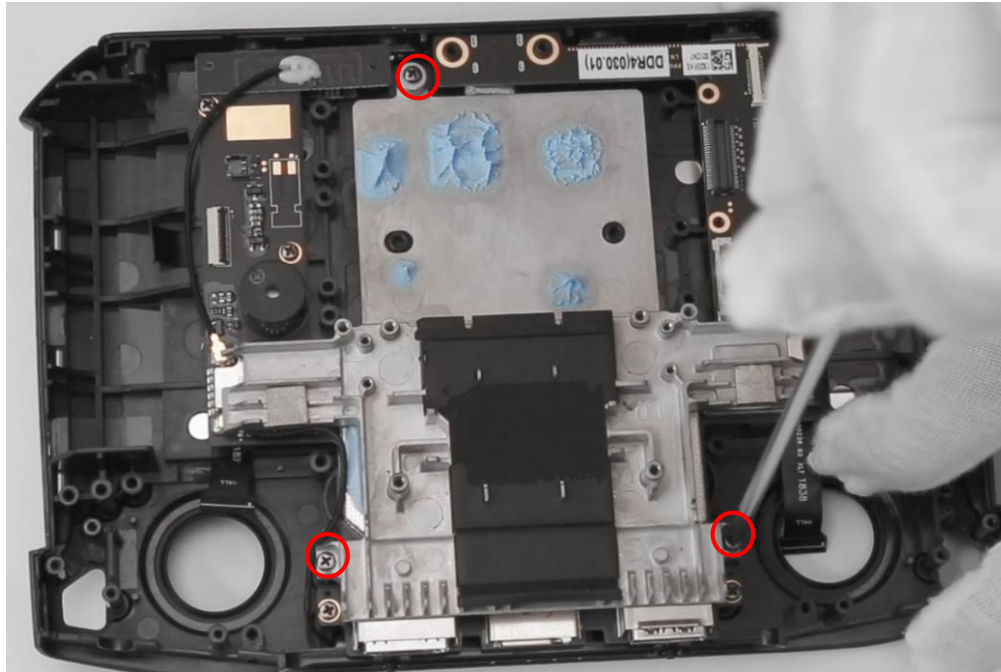
Pull out the fan module's flat cable. Unscrew the two screws securing the fan module, then remove the fan module.

Unscrew the two screws securing the main board's mounting piece, remove the mounting piece, and then disconnect the SDR board's flat cable.

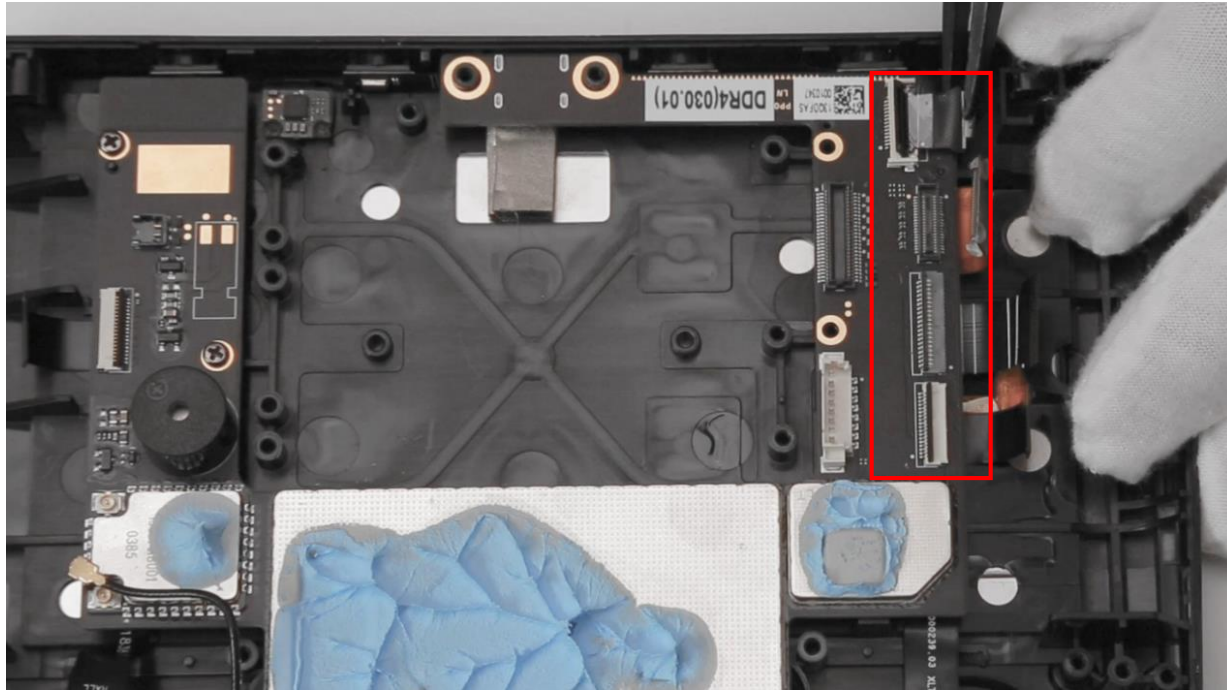


Disconnect the flat cable that connects the main board with the left Hall sensor board and the speaker's flat cable.
Note: UV glue is applied on the speaker's flat cable. Heat the UV glue with a heat gun (80°C, 2 minutes) and clean off the UV glue.

Unscrew the two screws securing the SDR board, then remove the SDR board.



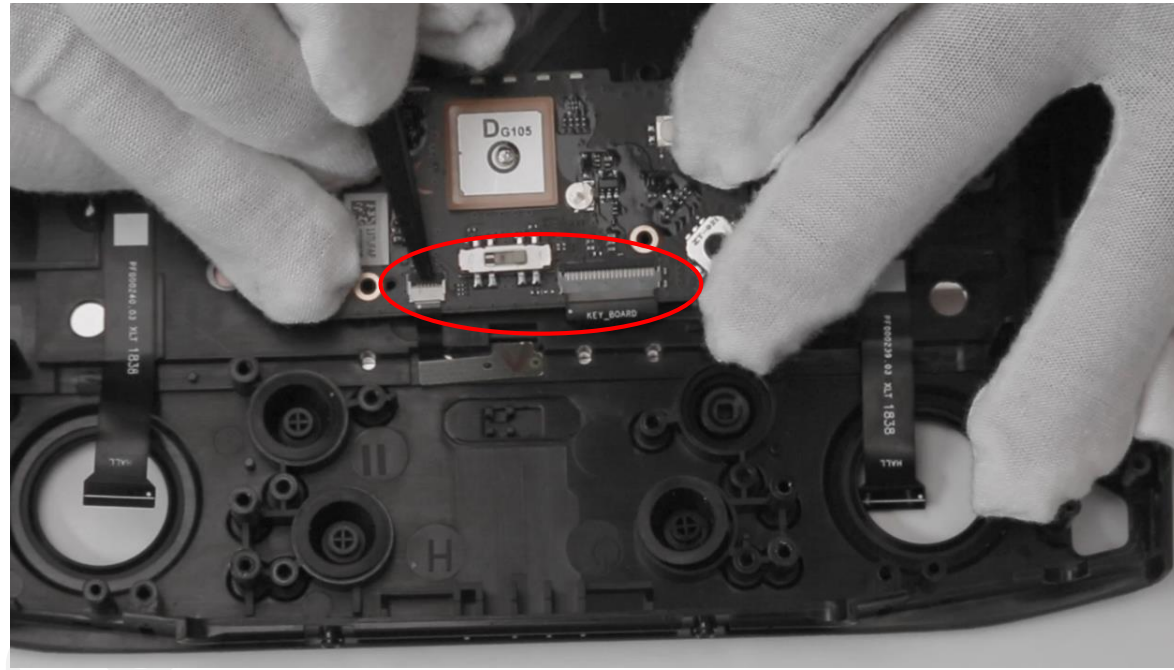
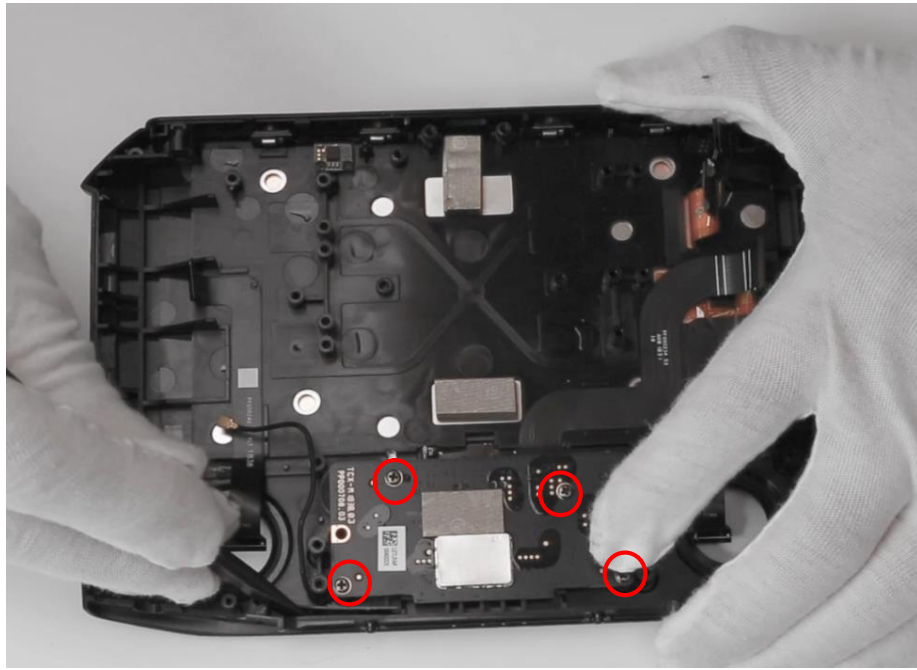
Unscrew the three screws securing the heat sink module, then remove the heat sink module. Disconnect the Wi-Fi antenna 1 and Wi-Fi antenna 2 connectors, and then remove the Wi-Fi Antenna 2.



Disconnect the flat cable that connects the main board with the right Hall sensor board, the flat cable that connects the main board with the combination buttons board, the flat cable that connects the main board with the screen, and the flat cable that connects the main board with the MIC board.

Note: UV glue is applied on the flat cable that connects the main board and the screen. Heat the UV glue with a heat gun (80°C, 2 minutes) and clean off the UV glue.

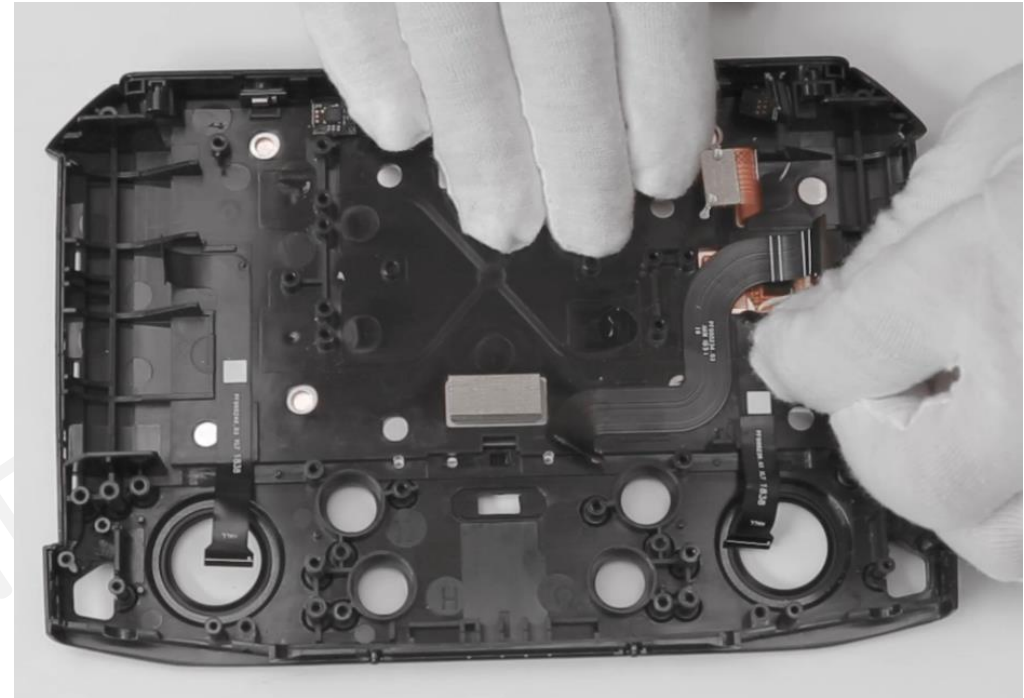
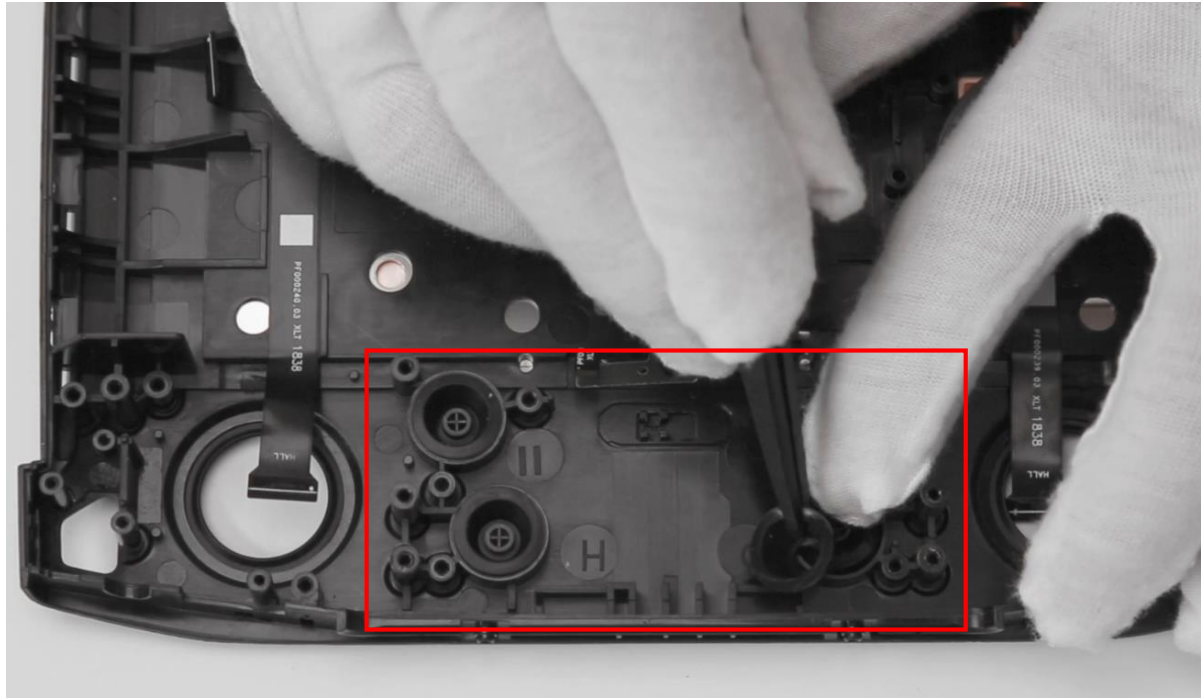
Unscrew the four screws securing the main board, and then remove the main board.



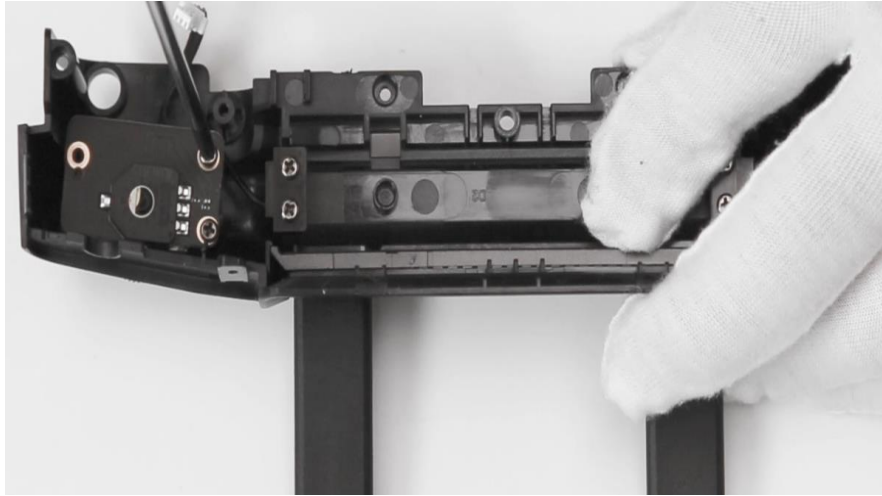
Remove the Wi-Fi Antenna 1, and then unscrew the four screws securing the combination buttons board.

Disconnect the photosensitive board's flat cable, and then disconnect the flat cable that connects the main board with the combination button board.

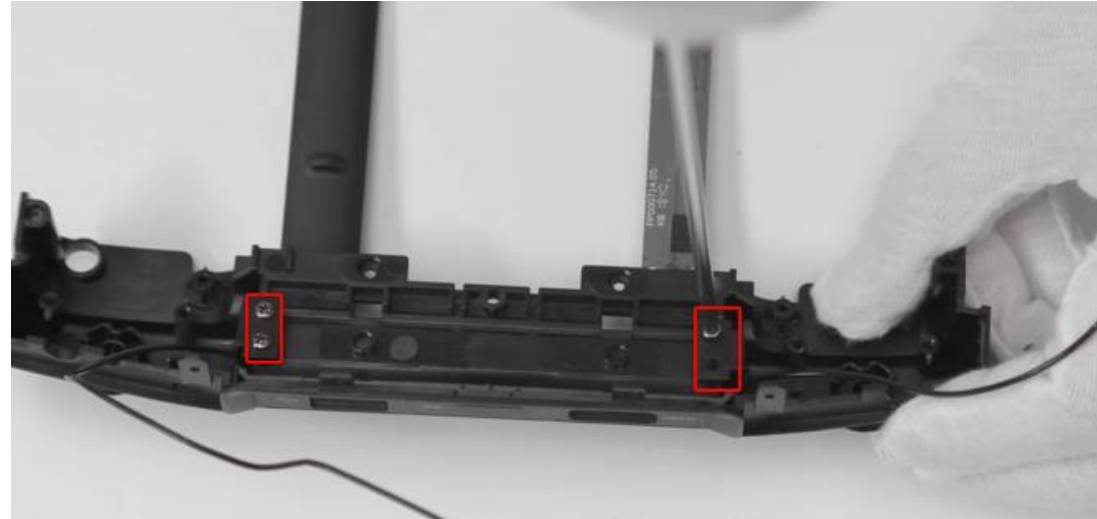
RM500 Smart Controller Disassembly



Remove the Power Button, RTH Button, Customizable Button, 5D Button and Flight Mode Switch on the front shell. Remove the photosensitive board in the middle of the upper shell and the bottom MIC board.



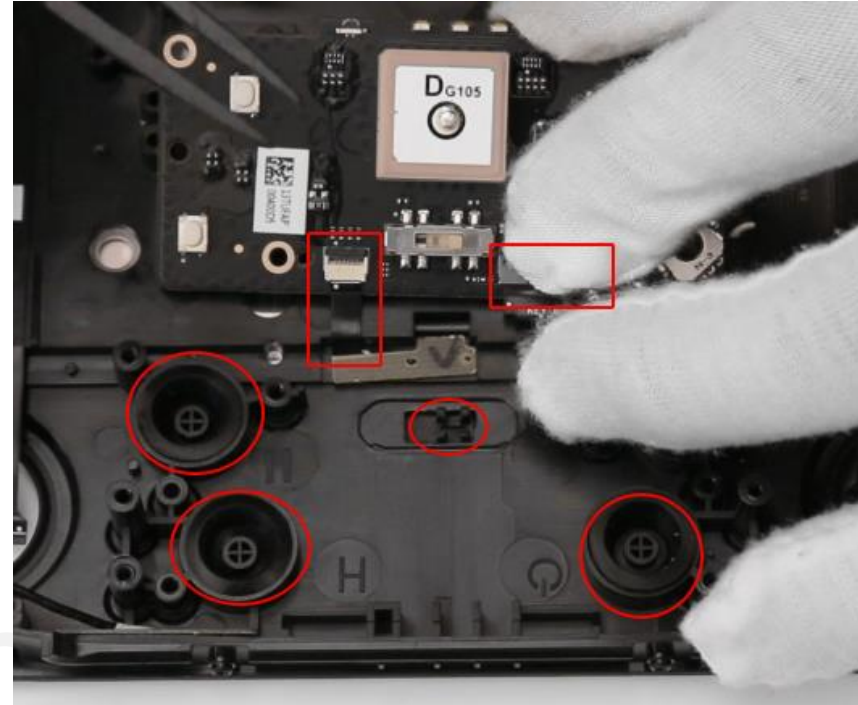
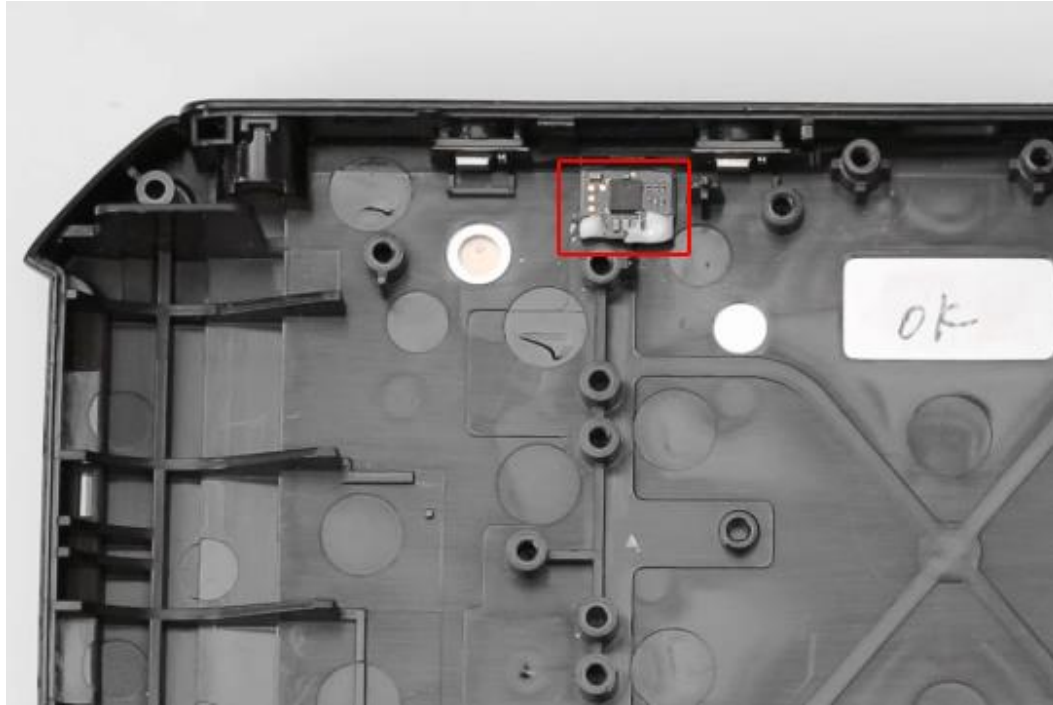
Unscrew the six screws securing the left and right dial boards, then remove the right dial board and right dial module.
Similarly, remove the left dial board and left dial module.



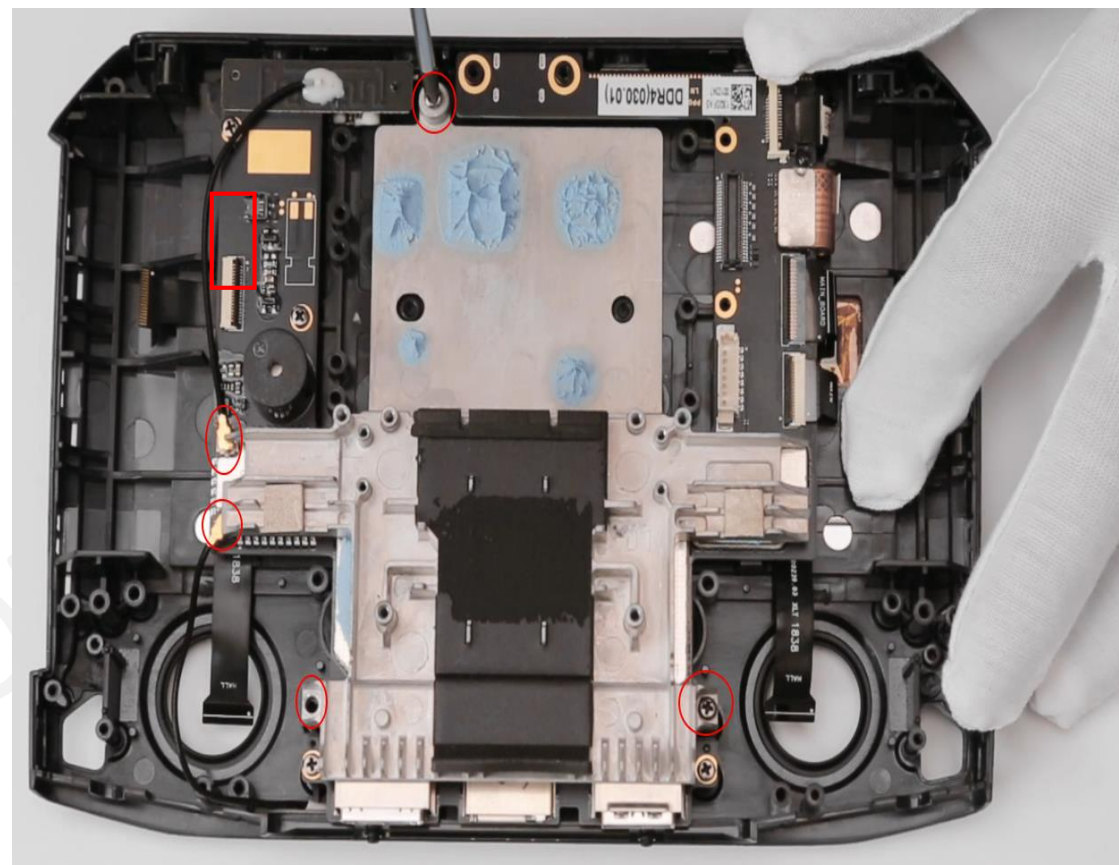
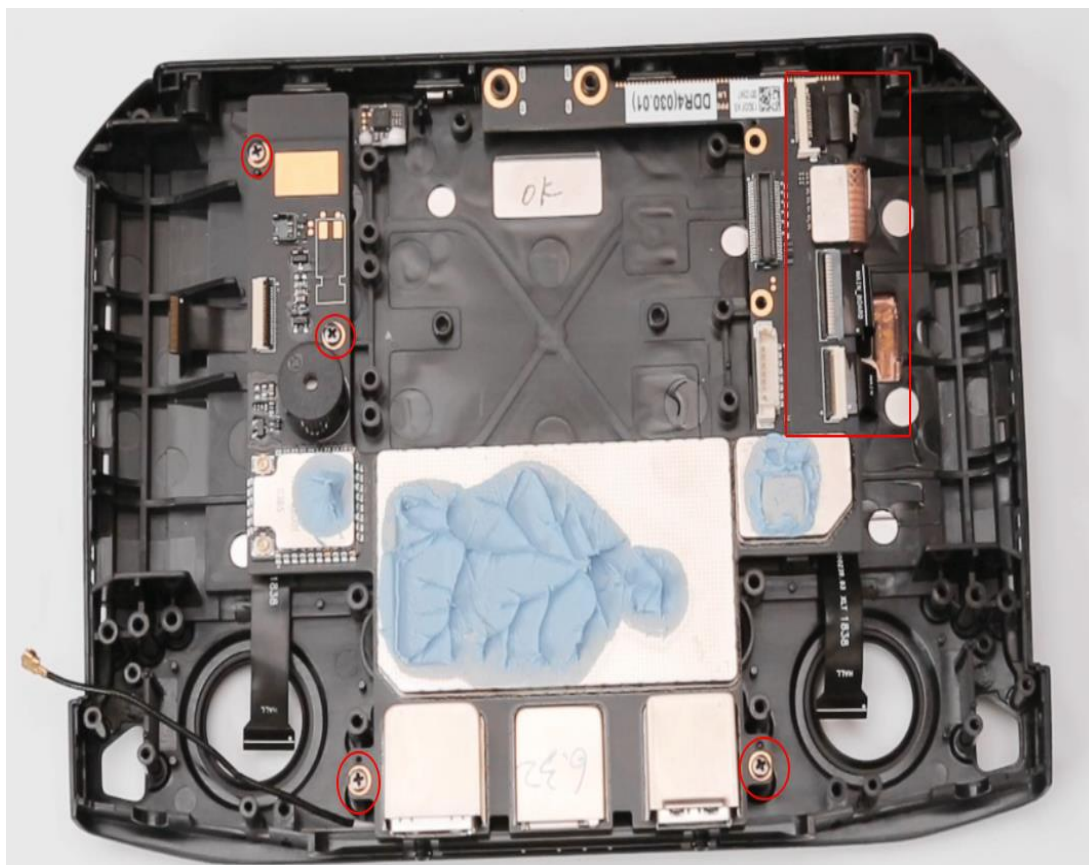
Heat the joint between the antenna's sleeve and antenna base with a heat gun. Heat one side of the sleeve. Be sure not to burn the antenna's base module. Remove the antenna's sleeve.

Unscrew the four screws securing the antenna rotation shaft. Then remove the antenna bracket and the left and right image transmission antenna.

RM500 Smart Controller Assembly

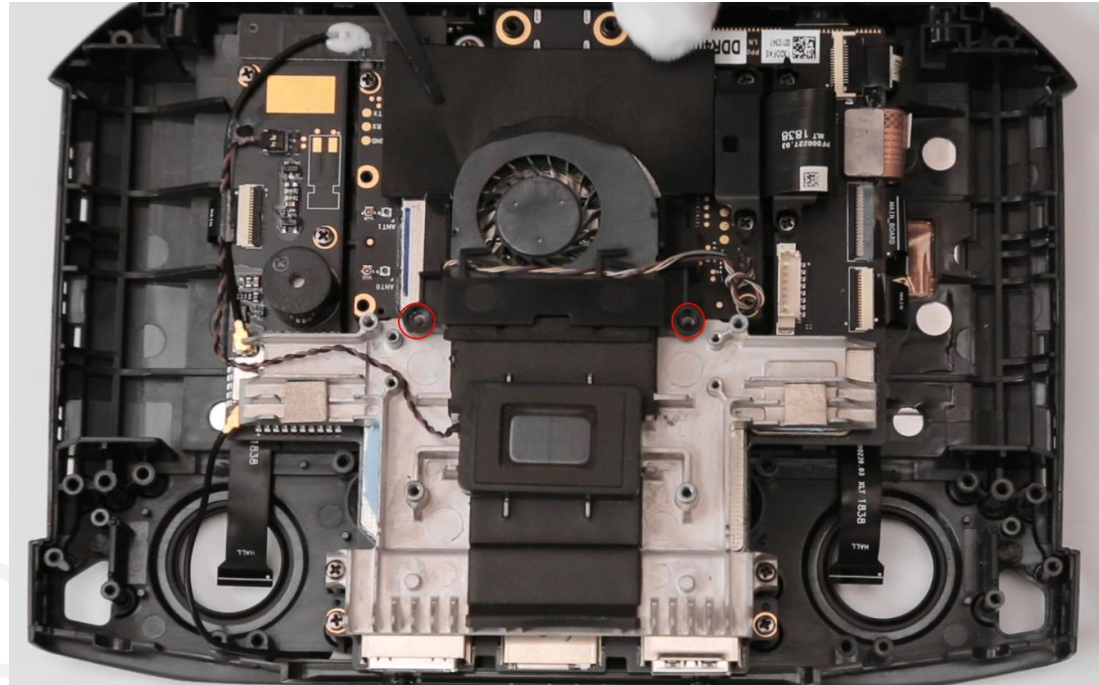
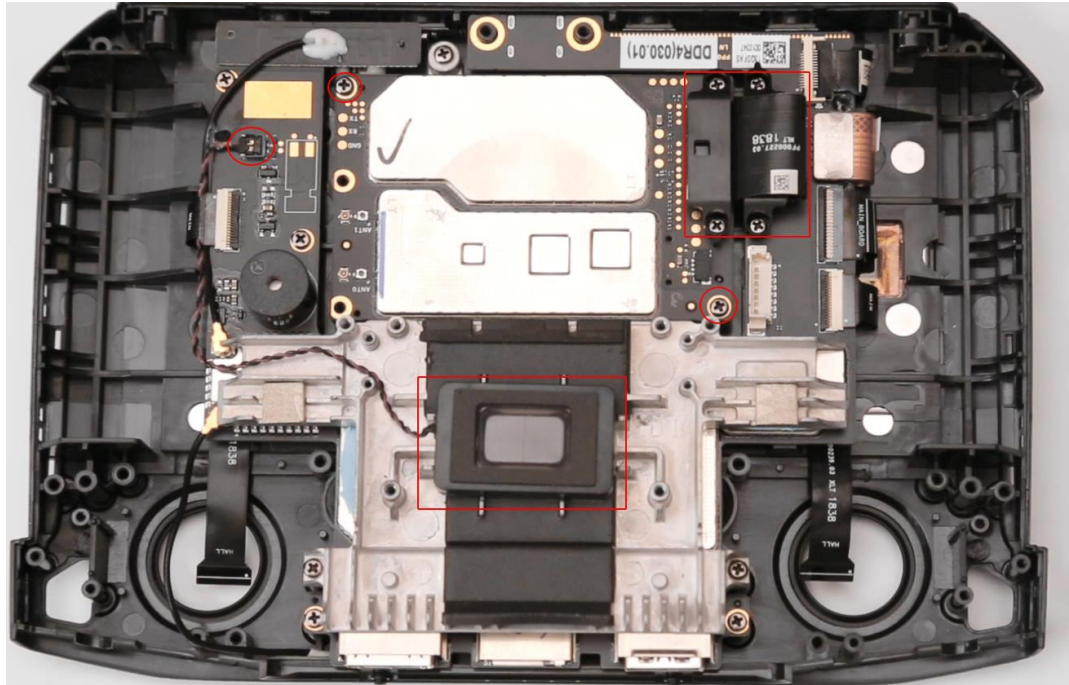


Mount the MIC board module, and then apply white glue to the positioning points. Attach the Wi-Fi Antenna 1. Mount the combination buttons board. The corresponding button and the connection cable should be mounted and connected in the right position.



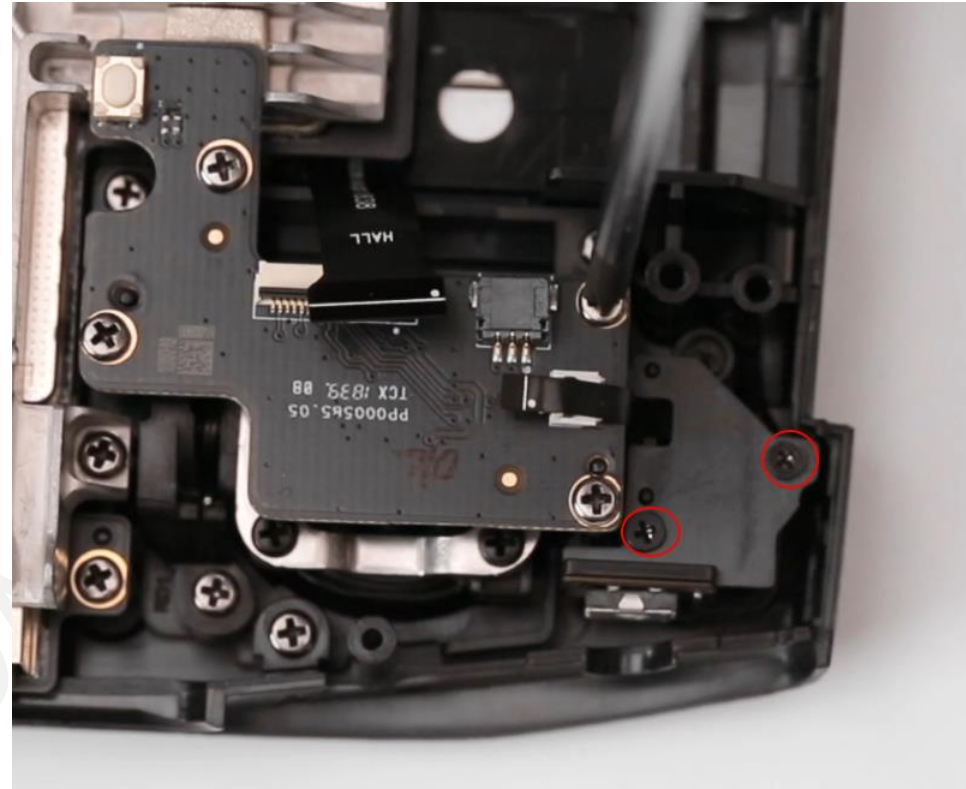
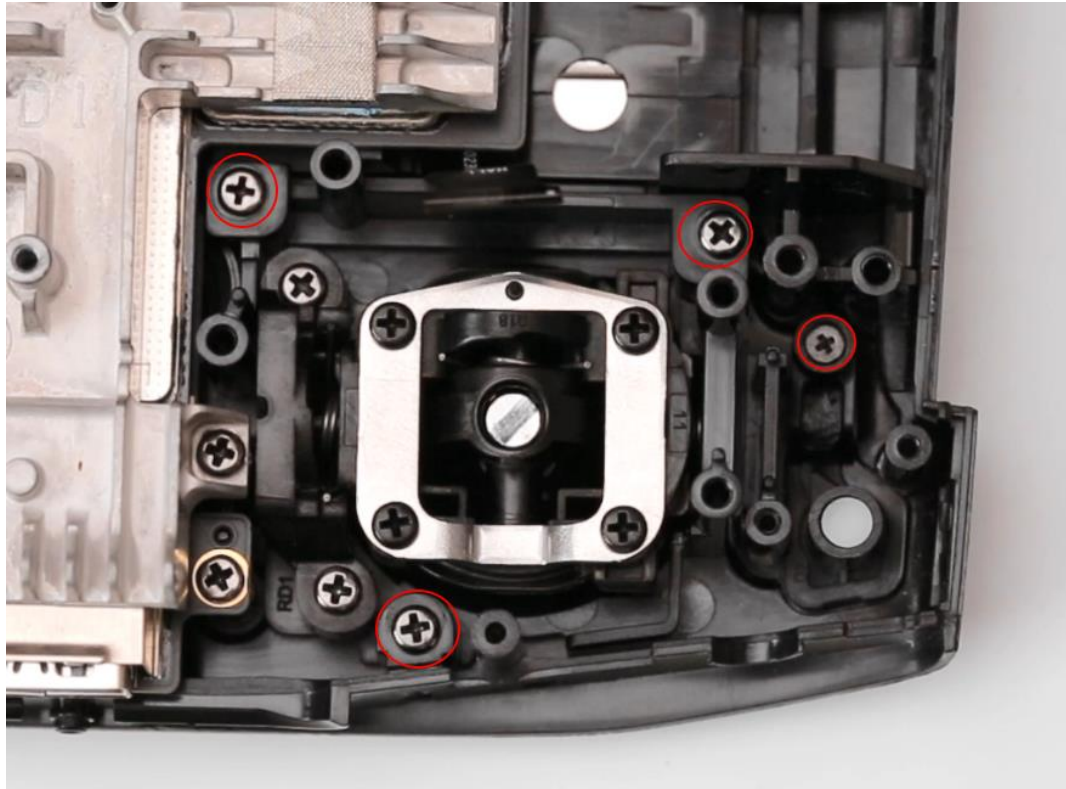
Mount the remote controller main board, screwing in the corresponding four screws. Connect the main board's cables and antennas.

Apply UV glue on both ends of the screen flat cable, then expose the glue to the UV light for ten seconds until it is solidified. Mount the heat sink module then secure it with the three screws.

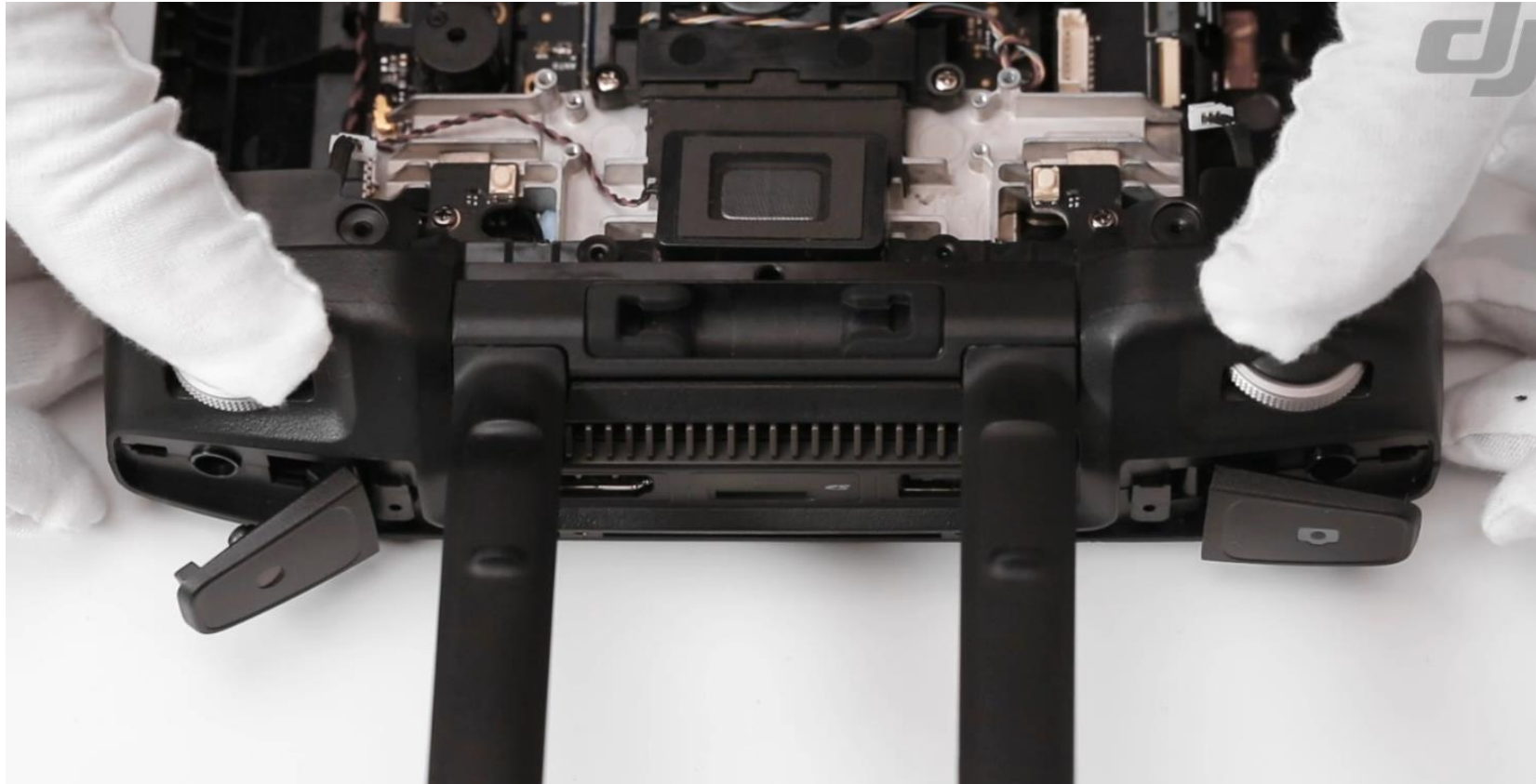


Mount the SDR board. Connect the flat cable that connects the SDR board with the remote controller main board. Screw the corresponding six screws to secure the SDR board and the flat cable.

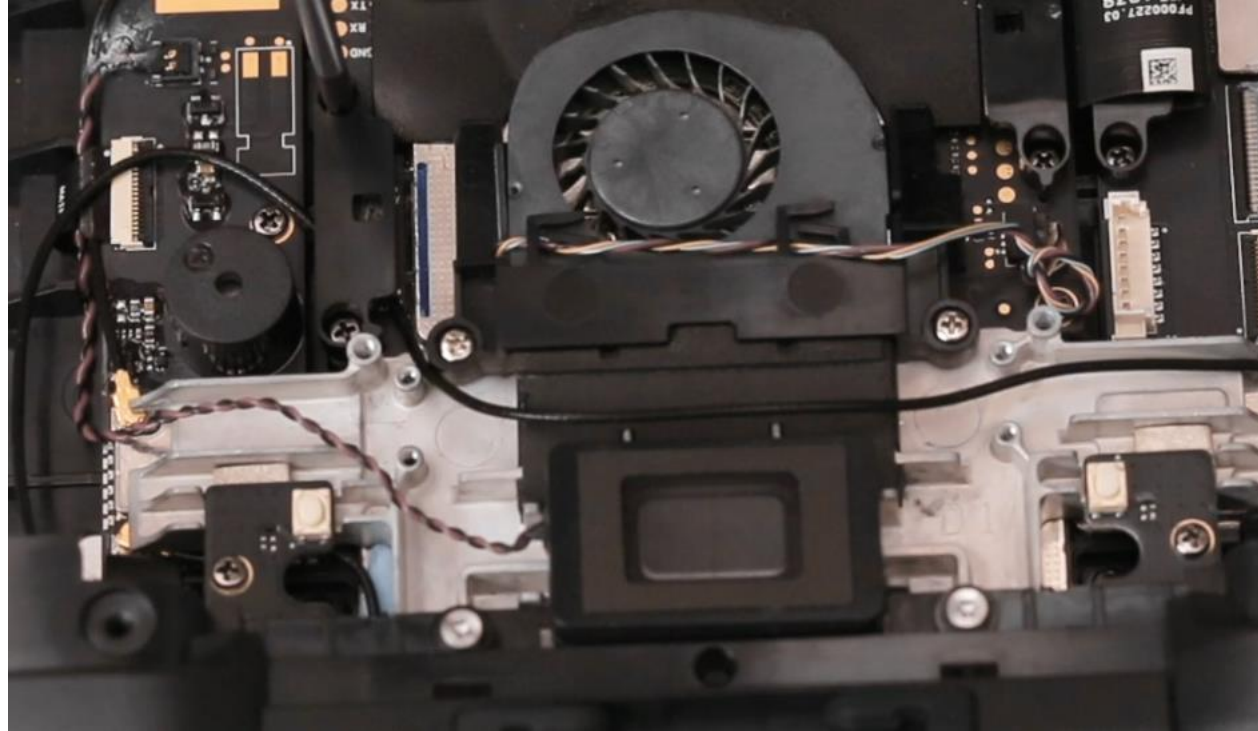
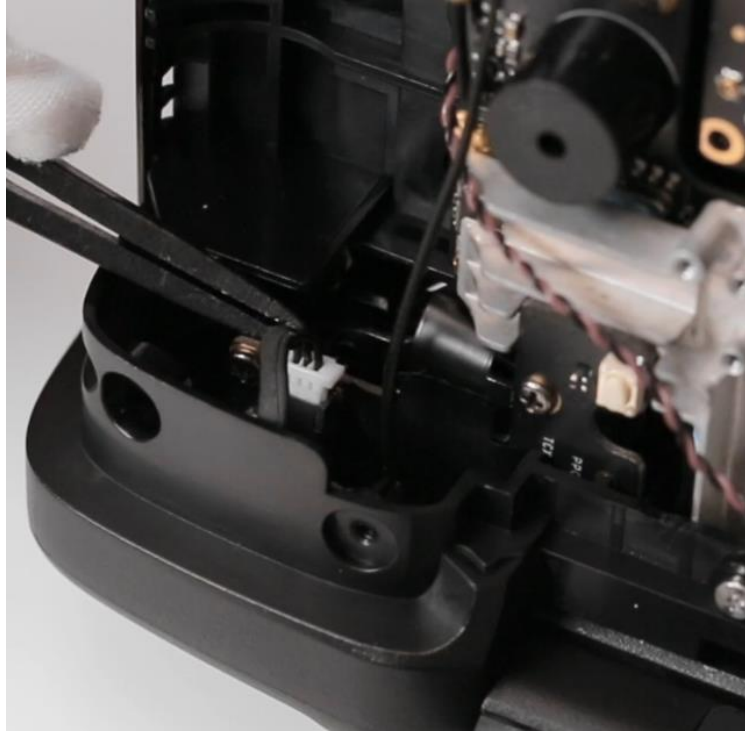
Mount the speaker and fan. Connect the corresponding cable and attach the SDR board's decorative piece.



Mount the right remote controller component and the Confirm Button, and then screw in the corresponding four screws.
Mount the FPV/Map Switch board and the right Hall sensor board. Tighten the corresponding six screws.
Mount the Back Button/Function Button, left control stick module, spraying board and left Hall component.

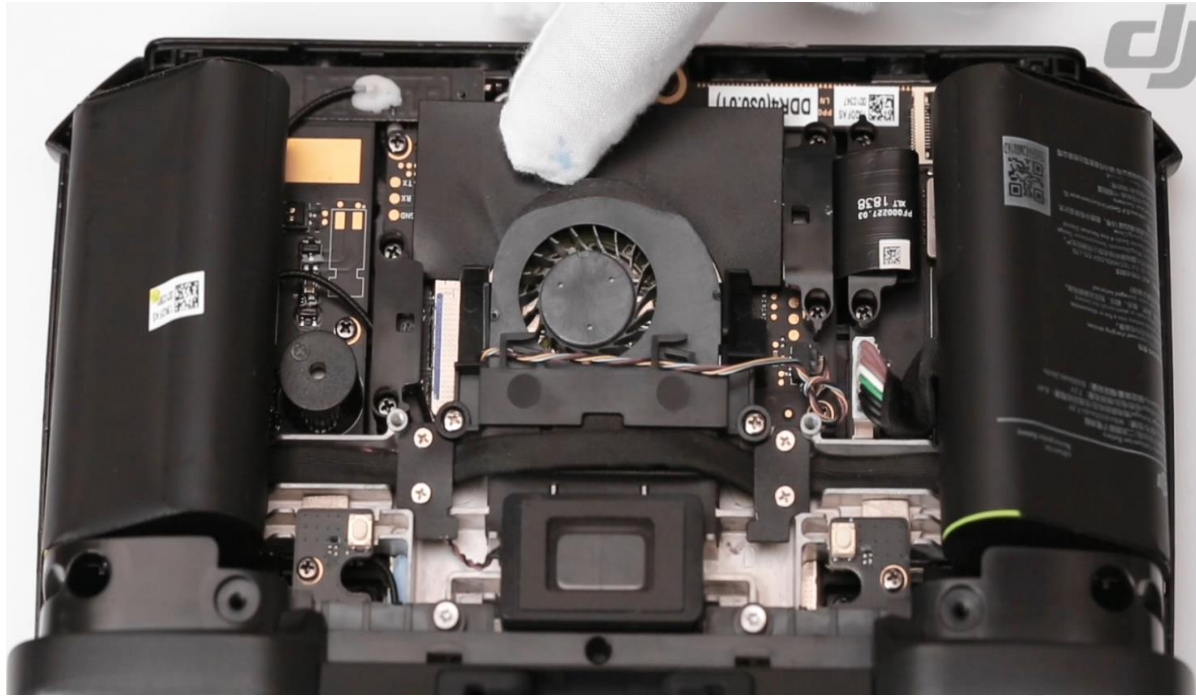


Mount the Spray Button and FPV/Map Switch Button, and then mount the front cover module.
Attach the spring to the FPV/Map Switch Button, and then tighten the eight screws securing the front cover module.
Mount the front decorative cover.



Attach the left and right dials' flat cables.

Attach the left antenna to the ANT0 port and the right antenna to the ANT1 port. Mount the image transmission antenna mounting piece and secure it with the two screws.



Mount the left and right batteries and insert the power cable. Attach the left and right battery cables' mounting pieces and secure them with four screws.
Mount the lower shell module and secure it with the eight screws.



Mount the rear cover module and secure it with the two screws.
Apply hot glue on the left and right rubber pads, then attach the left and right rubber pads.
Mount the expansion board module and the RTK dongle to complete mounting.

部件	故障	原因分析	处理方案
遥控器	无法充电	1.检查遥控器内置电池是否损坏	更换遥控器内置电池
	无法开机	1.检查外置电池电量	更换扩展模块接口板
		2.检查电池排线松动	扣紧或更换排线
		3.检查内置电池	更换内置电池
		4.检查主板模块	更换主板
	无法连接调参	1.调参版本不匹配	在官网页面下载对应版本调参
		2.电脑驱动安装失败	重装调参
		3.交叉更换正常主板检测	更换主板
	无法对频	1.检查遥控器固件是否与天空端匹配	升级遥控器固件
		2.交叉更换SDR板检测	更换SDR板
		3.交叉更换正常主板检测	更换主板
	规定范围内失联	1.检查天线是否损坏	更换天线
		2.交叉更换SDR板检测	更换SDR板
		3.交叉更换正常主板检测	更换主板
屏幕异常	检查屏幕或同轴线是否损坏	更换屏幕	
遥控器搜星异常	交叉跟换遥控器屏幕测试	更换组合按键板	
无法连接WiFi	1.检查WiFi天线是否损坏或未接好	扣紧或更换WiFi天线	
	2.交叉更换正常主板检测	更换主板	
波轮、按键失灵	检查转接板是否损坏	更换霍尔板/组合按键板/主板	
扩展模块	无法连接电池	遥控器主板Type-C接口损坏	更换主板
		扩展模块Type-C接口损坏	更换扩展模块接口板
	无法连接4G网卡	遥控器主板Type-C接口损坏	更换主板
		扩展模块Type-C接口损坏	更换扩展模块接口板
RTK Dongle	RTK搜星异常	1.检查RTK接口	扣紧或更换RTK板
		2.检查遥控器主板接口	更换遥控器主板